tropical: SCC

2026-10-24T16:13:39

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `0ea3d4b19330`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.59	[0.527, 0.655]	0.0825	5	0
adaptive	pointer	0.01	0.5576	[0.527, 0.578]	0.0343	5	0
adaptive	pointer	0.1	0.551	[0.526, 0.576]	0.0332	5	0
adaptive	pointer	1	0.5591	[0.528, 0.585]	0.0384	5	0
adaptive	pointer	10	0.5589	[0.526, 0.585]	0.0386	5	0
adaptive	pointer	100	0.5569	[0.523, 0.584]	0.0403	5	0
adaptive	pointer	1000	0.5574	[0.524, 0.58]	0.0363	5	0
adaptive	pointer	1e+04	0.559	[0.526, 0.595]	0.0476	5	0
adaptive	seq2seq	0	0.6517	[0.592, 0.711]	0.0803	5	0
adaptive	seq2seq	0.01	0.6666	[0.631, 0.701]	0.0458	5	0
adaptive	seq2seq	0.1	0.6624	[0.633, 0.689]	0.036	5	0
adaptive	seq2seq	1	0.6561	[0.621, 0.691]	0.0456	5	0
adaptive	seq2seq	10	0.6835	[0.668, 0.699]	0.0216	5	0
adaptive	seq2seq	100	0.6528	[0.626, 0.684]	0.0377	5	0
adaptive	seq2seq	1000	0.6553	[0.644, 0.674]	0.0207	5	0
adaptive	seq2seq	1e+04	0.69	[0.63, 0.757]	0.0853	5	0
tropical	pointer	0	0.6311	[0.567, 0.693]	0.0832	5	0
tropical	pointer	0.01	0.6806	[0.602, 0.775]	0.108	5	0
tropical	pointer	0.1	0.6838	[0.609, 0.758]	0.0996	5	0
tropical	pointer	1	0.6894	[0.663, 0.716]	0.0363	5	0
tropical	pointer	10	0.7018	[0.641, 0.775]	0.0865	5	0
tropical	pointer	100	0.6278	[0.567, 0.689]	0.0834	5	0
tropical	pointer	1000	0.6266	[0.565, 0.69]	0.0822	5	0
tropical	pointer	1e+04	0.6998	[0.607, 0.76]	0.104	5	0
tropical	seq2seq	0	0.6556	[0.612, 0.699]	0.0568	5	0
tropical	seq2seq	0.01	0.707	[0.659, 0.751]	0.0592	5	0
tropical	seq2seq	0.1	0.6884	[0.641, 0.736]	0.0662	5	0
tropical	seq2seq	1	0.761	[0.656, 0.866]	0.135	5	0
tropical	seq2seq	10	0.7688	[0.704, 0.838]	0.0898	5	0
tropical	seq2seq	100	0.7021	[0.678, 0.728]	0.0325	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.7326	[0.703, 0.762]	0.0385	5	0
tropical	seq2seq	1e+04	0.7196	[0.679, 0.76]	0.0525	5	0
vanilla	pointer	0	0.5676	[0.543, 0.592]	0.0319	5	0
vanilla	pointer	0.01	0.5763	[0.552, 0.593]	0.0272	5	0
vanilla	pointer	0.1	0.5733	[0.556, 0.59]	0.0229	5	0
vanilla	pointer	1	0.5735	[0.557, 0.588]	0.0207	5	0
vanilla	pointer	10	0.5732	[0.555, 0.594]	0.0252	5	0
vanilla	pointer	100	0.5799	[0.557, 0.599]	0.0262	5	0
vanilla	pointer	1000	0.5702	[0.55, 0.586]	0.0237	5	0
vanilla	pointer	1e+04	0.5736	[0.561, 0.586]	0.0163	5	0
vanilla	seq2seq	0	0.7089	[0.641, 0.776]	0.0859	5	0
vanilla	seq2seq	0.01	0.6787	[0.638, 0.719]	0.0523	5	0
vanilla	seq2seq	0.1	0.6901	[0.646, 0.734]	0.0592	5	0
vanilla	seq2seq	1	0.6831	[0.671, 0.692]	0.0135	5	0
vanilla	seq2seq	10	0.6781	[0.648, 0.708]	0.0404	5	0
vanilla	seq2seq	100	0.6474	[0.633, 0.666]	0.0203	5	0
vanilla	seq2seq	1000	0.6742	[0.656, 0.692]	0.0226	5	0
vanilla	seq2seq	1e+04	0.6705	[0.623, 0.718]	0.0624	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.03248	[-0.114, 0.0486]	0.7910
adaptive	pointer	0.1	-0.03902	[-0.12, 0.0421]	0.6383
adaptive	pointer	1	-0.03098	[-0.112, 0.0501]	0.8227
adaptive	pointer	10	-0.03115	[-0.112, 0.0499]	0.8191
adaptive	pointer	100	-0.03315	[-0.114, 0.048]	0.7764
adaptive	pointer	1000	-0.0326	[-0.114, 0.0485]	0.7885
adaptive	pointer	1e+04	-0.03108	[-0.112, 0.05]	0.8208
adaptive	seq2seq	0.01	0.01491	[-0.0757, 0.106]	0.9974
adaptive	seq2seq	0.1	0.01067	[-0.0799, 0.101]	0.9997
adaptive	seq2seq	1	0.00445	[-0.0861, 0.095]	1.0000
adaptive	seq2seq	10	0.03182	[-0.0588, 0.122]	0.8705
adaptive	seq2seq	100	0.001083	[-0.0895, 0.0917]	1.0000
adaptive	seq2seq	1000	0.003657	[-0.0869, 0.0943]	1.0000
adaptive	seq2seq	1e+04	0.03836	[-0.0522, 0.129]	0.7495
tropical	pointer	0.01	0.04958	[-0.104, 0.203]	0.9078
tropical	pointer	0.1	0.0527	[-0.101, 0.206]	0.8815
tropical	pointer	1	0.0583	[-0.0951, 0.212]	0.8261
tropical	pointer	10	0.07076	[-0.0827, 0.224]	0.6773
tropical	pointer	100	-0.003289	[-0.157, 0.15]	1.0000
tropical	pointer	1000	-0.004432	[-0.158, 0.149]	1.0000
tropical	pointer	1e+04	0.06873	[-0.0847, 0.222]	0.7031
tropical	seq2seq	0.01	0.05141	[-0.0763, 0.179]	0.7874
tropical	seq2seq	0.1	0.03281	[-0.0949, 0.161]	0.9682
tropical	seq2seq	1	0.1054	[-0.0223, 0.233]	0.1403
tropical	seq2seq	10	0.1132	[-0.0145, 0.241]	0.0993
tropical	seq2seq	100	0.04652	[-0.0812, 0.174]	0.8510
tropical	seq2seq	1000	0.07698	[-0.0507, 0.205]	0.4111
tropical	seq2seq	1e+04	0.06401	[-0.0637, 0.192]	0.5988
vanilla	pointer	0.01	0.008738	[-0.0343, 0.0518]	0.9910
vanilla	pointer	0.1	0.005685	[-0.0374, 0.0488]	0.9993
vanilla	pointer	1	0.005887	[-0.0372, 0.049]	0.9992

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	10	0.005638	[-0.0374, 0.0487]	0.9994
vanilla	pointer	100	0.01227	[-0.0308, 0.0553]	0.9475
vanilla	pointer	1000	0.002595	[-0.0405, 0.0457]	1.0000
vanilla	pointer	1e+04	0.00599	[-0.0371, 0.0491]	0.9991
vanilla	seq2seq	0.01	-0.0302	[-0.118, 0.0577]	0.8812
vanilla	seq2seq	0.1	-0.01878	[-0.107, 0.0691]	0.9880
vanilla	seq2seq	1	-0.02575	[-0.114, 0.0621]	0.9400
vanilla	seq2seq	10	-0.03082	[-0.119, 0.057]	0.8712
vanilla	seq2seq	100	-0.06145	[-0.149, 0.0264]	0.2677
vanilla	seq2seq	1000	-0.03469	[-0.123, 0.0532]	0.8010
vanilla	seq2seq	1e+04	-0.03844	[-0.126, 0.0494]	0.7232

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	5.316×10^{-4}	[-0.00251, 0.00331]	0.8611	-0.0171	0.9126
adaptive	seq2seq	0.001892	[-0.00862, 0.0124]	0.6205	-0.0248	0.8706
tropical	pointer	-0.004228	[-0.0199, 0.0115]	0.5675	-0.0648	0.6431
tropical	seq2seq	0.002403	[-0.00689, 0.0123]	0.7064	0.0857	0.5161
vanilla	pointer	-2.872×10^{-4}	[-0.00203, 0.00117]	0.8769	-0.0514	0.7121
vanilla	seq2seq	-0.003294	[-0.00899, 0.00167]	0.3486	-0.0705	0.6189

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.06346	[-0.0621, 0.189]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.1043	[-0.0451, 0.254]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.1105	[-0.0326, 0.254]	0.9873
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1159	[0.0545, 0.177]	0.1014
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1286	[0.0038, 0.253]	0.5964
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04791	[-0.082, 0.178]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.05644	[-0.0585, 0.171]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.1262	[-0.0144, 0.267]	0.8073
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.05329	[-0.197, 0.0904]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.02831	[-0.0797, 0.136]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.0017	[-0.0968, 0.0934]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.07784	[-0.0814, 0.237]	1.0000

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.09075	[-0.0214, 0.203]	0.9688
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.05468	[0.0072, 0.102]	0.4948
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.05837	[0.00476, 0.112]	0.5466
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.04916	[-0.0574, 0.156]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.02244	[-0.0739, 0.119]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.01877	[-0.0382, 6.41×10^{-4}]	0.8244
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.02226	[-0.0409, -0.00361]	0.4727
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.01442	[-0.0406, 0.0117]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.01435	[-0.0369, 0.00818]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.02297	[-0.0572, 0.0113]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.01275	[-0.0415, 0.016]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.01463	[-0.0619, 0.0327]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.05721	[-0.145, 0.0309]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.0121	[-0.0868, 0.0626]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.02776	[-0.126, 0.071]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.027	[-0.0955, 0.0415]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.005433	[-0.0652, 0.0761]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.005327	[-0.0557, 0.0663]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.01886	[-0.0671, 0.0294]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.01959	[-0.0899, 0.129]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.06164	[-0.18, 0.0567]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.109	[-0.186, -0.0324]	0.2621
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.1113	[-0.172, -0.0508]	0.1318
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.09707	[-0.178, -0.0161]	0.3495
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.1246	[-0.18, -0.0693]	0.0765
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.09587	[-0.144, -0.0475]	0.1115
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.0979	[-0.145, -0.0507]	0.0992
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.1311	[-0.207, -0.0553]	0.1553

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.02454	[-0.162, 0.113]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.02637	[-0.0993, 0.0465]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.004646	[-0.109, 0.0994]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.07162	[-0.235, 0.0913]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.06699	[-0.281, 0.147]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.07435	[-0.173, 0.0241]	0.8332	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.1059	[-0.216, 0.00415]	0.5013	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.01981	[-0.142, 0.102]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.1413	[-0.252, -0.0301]	0.3394	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.1024	[-0.193, -0.0121]	0.3807	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.1168	[-0.194, -0.0401]	0.2281	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.1096	[-0.146, -0.0729]	0.0278	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.1048	[-0.178, -0.0317]	0.2621	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.06758	[-0.121, -0.0137]	0.3394	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.104	[-0.158, -0.0503]	0.1154	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.09686	[-0.183, -0.0103]	0.3807	

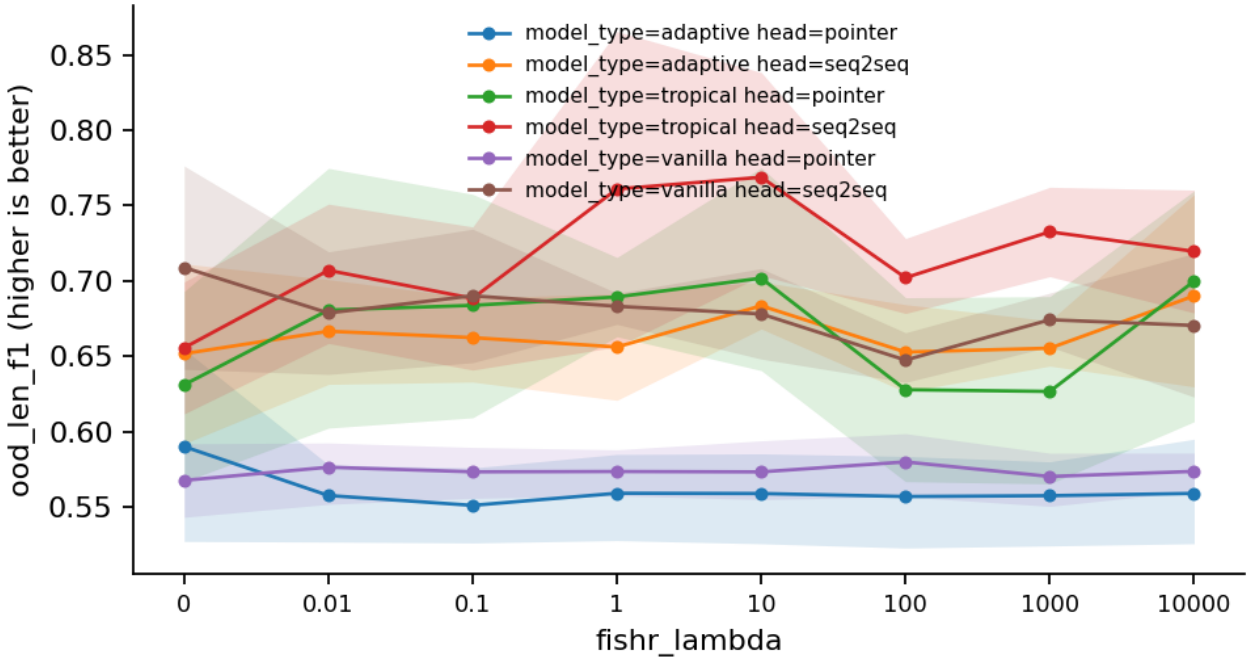


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

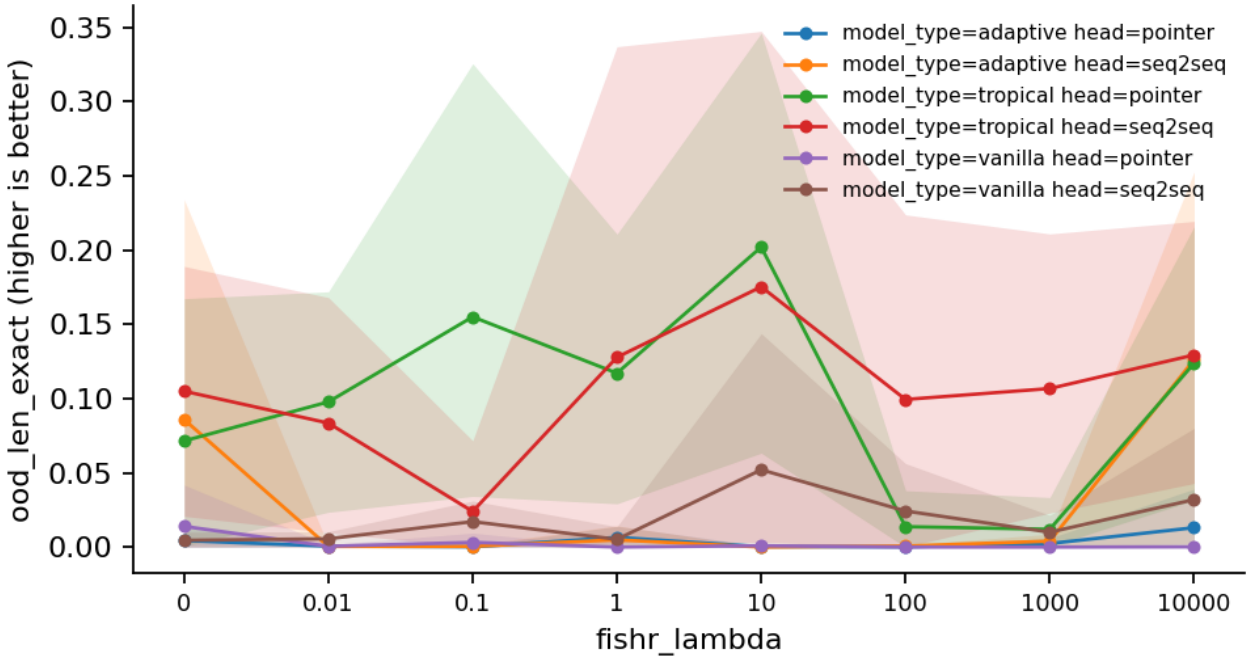


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.00435	[0, 0.0131]	0.00973	5	0
adaptive	pointer	0.01	6.500×10^{-4}	[0, 0.00145]	9.29×10^{-4}	5	0
adaptive	pointer	0.1	1.500×10^{-4}	[0, 3.50×10^{-4}]	2.24×10^{-4}	5	0
adaptive	pointer	1	0.00655	[0, 0.014]	0.00912	5	0
adaptive	pointer	10	3.000×10^{-4}	[0, 9.00×10^{-4}]	6.71×10^{-4}	5	0
adaptive	pointer	100	1.500×10^{-4}	[0, 4.50×10^{-4}]	3.35×10^{-4}	5	0
adaptive	pointer	1000	0.0024	[0, 0.006]	0.00391	5	0
adaptive	pointer	1e+04	0.01305	[0, 0.0392]	0.0292	5	0
adaptive	seq2seq	0	0.0859	[2.00×10^{-4}, 0.234]	0.161	5	0
adaptive	seq2seq	0.01	4.000×10^{-4}	[0, 8.00×10^{-4}]	5.48×10^{-4}	5	0
adaptive	seq2seq	0.1	5.000×10^{-4}	[0, 0.0014]	9.84×10^{-4}	5	0
adaptive	seq2seq	1	0.0049	[0, 0.0144]	0.0107	5	0
adaptive	seq2seq	10	0	[0, 0]	0	5	0
adaptive	seq2seq	100	7.000×10^{-4}	[0, 0.0014]	9.59×10^{-4}	5	0
adaptive	seq2seq	1000	0.004	[5.00×10^{-5}, 0.00875]	0.00557	5	0
adaptive	seq2seq	1e+04	0.1261	[5.00×10^{-4}, 0.253]	0.172	5	0
tropical	pointer	0	0.0715	[6.50×10^{-4}, 0.167]	0.12	5	0
tropical	pointer	0.01	0.09775	[0.0236, 0.172]	0.0949	5	0
tropical	pointer	0.1	0.155	[0.0342, 0.326]	0.196	5	0
tropical	pointer	1	0.1171	[0.0292, 0.211]	0.116	5	0
tropical	pointer	10	0.202	[0.0634, 0.346]	0.185	5	0
tropical	pointer	100	0.01375	[0, 0.0381]	0.0265	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.012	[4.50×10^{-4}, 0.0335]	0.0239	5	0
tropical	pointer	1e+04	0.1237	[0.0316, 0.216]	0.12	5	0
tropical	seq2seq	0	0.105	[0.0209, 0.189]	0.106	5	0
tropical	seq2seq	0.01	0.0835	[0.00975, 0.168]	0.101	5	0
tropical	seq2seq	0.1	0.024	[0, 0.0716]	0.0531	5	0
tropical	seq2seq	1	0.1278	[0.0028, 0.337]	0.228	5	0
tropical	seq2seq	10	0.1752	[0.00325, 0.347]	0.233	5	0
tropical	seq2seq	100	0.0994	[1.50×10^{-4}, 0.224]	0.143	5	0
tropical	seq2seq	1000	0.1067	[0.023, 0.211]	0.118	5	0
tropical	seq2seq	1e+04	0.1294	[0.043, 0.219]	0.117	5	0
vanilla	pointer	0	0.01395	[0, 0.0418]	0.0311	5	0
vanilla	pointer	0.01	5.500×10^{-4}	[0, 0.00165]	0.00123	5	0
vanilla	pointer	0.1	0.00315	[0, 0.00945]	0.00704	5	0
vanilla	pointer	1	0	[0, 0]	0	5	0
vanilla	pointer	10	9.000×10^{-4}	[0, 0.0027]	0.00201	5	0
vanilla	pointer	100	0	[0, 0]	0	5	0
vanilla	pointer	1000	0	[0, 0]	0	5	0
vanilla	pointer	1e+04	1.500×10^{-4}	[0, 4.50×10^{-4}]	3.35×10^{-4}	5	0
vanilla	seq2seq	0	0.00455	[5.00×10^{-5}, 0.0101]	0.00642	5	0
vanilla	seq2seq	0.01	0.0056	[7.50×10^{-4}, 0.0105]	0.00629	5	0
vanilla	seq2seq	0.1	0.01715	[0.0034, 0.0311]	0.0182	5	0
vanilla	seq2seq	1	0.0052	[1.00×10^{-4}, 0.0138]	0.00938	5	0
vanilla	seq2seq	10	0.05205	[0.0019, 0.144]	0.102	5	0
vanilla	seq2seq	100	0.0243	[0, 0.0563]	0.0406	5	0
vanilla	seq2seq	1000	0.01	[4.50×10^{-4}, 0.0215]	0.0134	5	0
vanilla	seq2seq	1e+04	0.03195	[0.00215, 0.08]	0.0521	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0	[-0.0128, 0.0014]	1.0000
adaptive	pointer	0.1	0	[-0.013, 3.01×10^{-4}]	1.0000
adaptive	pointer	1	0	[-0.0102, 0.014]	1.0000
adaptive	pointer	10	0	[-0.0131, 9.00×10^{-4}]	1.0000
adaptive	pointer	100	0	[-0.0131, 4.50×10^{-4}]	1.0000
adaptive	pointer	1000	0	[-0.0124, 0.0054]	1.0000
adaptive	pointer	1e+04	0	[-0.0131, 0.0392]	1.0000
adaptive	seq2seq	0.01	-0.001	[-0.233, 2.00×10^{-4}]	1.0000
adaptive	seq2seq	0.1	-0.001	[-0.233, -1.49×10^{-4}]	1.0000
adaptive	seq2seq	1	-0.001	[-0.229, 0.0096]	1.0000
adaptive	seq2seq	10	-0.001	[-0.234, -2.00×10^{-4}]	1.0000
adaptive	seq2seq	100	-0.001	[-0.233, -2.00×10^{-4}]	1.0000
adaptive	seq2seq	1000	-7.500×10^{-4}	[-0.225, -1.50×10^{-4}]	1.0000
adaptive	seq2seq	1e+04	5.000×10^{-4}	[-0.222, 0.229]	1.0000
tropical	pointer	0.01	0.004	[-0.0313, 0.0956]	1.0000
tropical	pointer	0.1	0.0825	[0.0255, 0.148]	0.8750
tropical	pointer	1	7.500×10^{-4}	[-0.00445, 0.102]	1.0000
tropical	pointer	10	0.0505	[0.0157, 0.281]	0.8750
tropical	pointer	100	-0.00325	[-0.144, 0.0032]	1.0000
tropical	pointer	1000	-0.00325	[-0.169, 6.00×10^{-4}]	1.0000
tropical	pointer	1e+04	0.03525	[-0.0185, 0.133]	1.0000
tropical	seq2seq	0.01	0	[-0.105, 0.0586]	1.0000

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
tropical	seq2seq	0.1	-0.1003	[-0.189, 0.0328]	1.0000
tropical	seq2seq	1	-0.099	[-0.184, 0.3]	1.0000
tropical	seq2seq	10	-0.0015	[-0.127, 0.288]	1.0000
tropical	seq2seq	100	-0.045	[-0.133, 0.165]	1.0000
tropical	seq2seq	1000	0.011	[-0.086, 0.0983]	1.0000
tropical	seq2seq	1e+04	0.03725	[-0.012, 0.062]	1.0000
vanilla	pointer	0.01	0	[-0.0417, 0.0016]	1.0000
vanilla	pointer	0.1	0	[-0.0323, 0]	1.0000
vanilla	pointer	1	0	[-0.0418, 0]	1.0000
vanilla	pointer	10	0	[-0.0391, 0]	1.0000
vanilla	pointer	100	0	[-0.0418, 0]	1.0000
vanilla	pointer	1000	0	[-0.0418, 0]	1.0000
vanilla	pointer	1e+04	0	[-0.0413, 0]	1.0000
vanilla	seq2seq	0.01	0	[-0.0031, 0.00625]	1.0000
vanilla	seq2seq	0.1	0.004	$[8.00 \times 10^{-4}, 0.0244]$	1.0000
vanilla	seq2seq	1	0	[-0.0075, 0.0131]	1.0000
vanilla	seq2seq	10	0.00825	[0.00165, 0.134]	1.0000
vanilla	seq2seq	100	0	[-0.0034, 0.0571]	1.0000
vanilla	seq2seq	1000	0.00225	$[3.50 \times 10^{-4}, 0.0135]$	1.0000
vanilla	seq2seq	1e+04	0.0045	[-0.0017, 0.0669]	1.0000

Table 7: `ood_len_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.06013	[-0.291, 0.171]	0.7315	-0.04	0.7249
adaptive	seq2seq	0.3337	[-0.0681, 0.729]	0.0501	0.238	0.0516
tropical	pointer	-0.1942	[-0.393, 0.0302]	0.2635	-0.133	0.3173
tropical	seq2seq	0.2526	[-0.0076, 0.526]	0.1432	0.185	0.1654
vanilla	pointer	-0.105	[-0.459, 0.0364]	0.5485	-0.0419	0.5745
vanilla	seq2seq	0.03393	[-0.234, 0.283]	0.8466	0.0286	0.8216

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.05755	[-0.108, 0.223]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0972	[-0.0214, 0.216]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.1518	[-0.0931, 0.397]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1171	[-0.0265, 0.261]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.2011	[-0.03, 0.432]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01375	[-0.0192, 0.0467]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.012	[-0.0177, 0.0417]	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.1236	[-0.0255, 0.273]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1004	[-0.0294, 0.23]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.0779	[-0.0466, 0.202]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.00685	[-0.0525, 0.0662]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1226	[-0.149, 0.395]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1232	[-0.236, 0.483]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.0751	[-0.0979, 0.248]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.09675	[-0.0597, 0.253]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.0974	[-0.09, 0.285]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.0096	[-0.0528, 0.0336]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	1.000×10^{-4}	[-0.00214, 0.00234]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.003	[-0.0115, 0.0055]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00655	[-0.00478, 0.0179]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-6.000×10^{-4}	[-0.00227, 0.00107]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	1.500×10^{-4}	$[-2.66 \times 10^{-4}, 5.66 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.0024	[-0.00246, 0.00726]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.0129	[-0.0229, 0.0487]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.08135	[-0.122, 0.284]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.0052	[-0.0131, 0.00265]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01665	[-0.0388, 0.00551]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-3.000×10^{-4}	[-0.0203, 0.0197]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.05205	[-0.179, 0.0745]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0236	[-0.0747, 0.0275]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.006	[-0.0227, 0.0107]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.0942	[-0.094, 0.282]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.08155	[-0.285, 0.122]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	2.500×10^{-4}	[-0.00142, 0.00192]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-3.500×10^{-4}	[-0.00171, 0.00101]	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.00165	[-0.019, 0.0223]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	3.000×10^{-4}	$[-5.33 \times 10^{-4}, 0.00113]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-5.500×10^{-4}	$[-0.00154, 4.41 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.0016	[-0.00895, 0.00575]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.1131	[-0.307, 0.0808]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.03345	[-0.146, 0.079]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.01425	[-0.112, 0.14]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.131	[-0.142, 0.404]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.0107	[-0.407, 0.385]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.02675	[-0.487, 0.54]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.08565	[-0.256, 0.0851]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.09475	[-0.252, 0.0621]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.00565	[-0.18, 0.168]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0094	[-0.0272, 0.046]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.00505	[-0.0137, 0.0036]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.014	[-0.0403, 0.0123]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.0052	[-0.0169, 0.00645]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.05115	[-0.178, 0.076]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.0243	[-0.0748, 0.0262]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.01	[-0.0266, 0.00659]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.0318	[-0.0966, 0.033]	1.0000

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9707	[0.969, 0.972]	0.00194	5	0
adaptive	pointer	0.01	0.971	[0.969, 0.973]	0.00261	5	0
adaptive	pointer	0.1	0.971	[0.969, 0.973]	0.00243	5	0
adaptive	pointer	1	0.9708	[0.969, 0.973]	0.00289	5	0
adaptive	pointer	10	0.9711	[0.969, 0.973]	0.00251	5	0
adaptive	pointer	100	0.9709	[0.969, 0.973]	0.00243	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	1000	0.9708	[0.969, 0.973]	0.00247	5	0
adaptive	pointer	1e+04	0.971	[0.969, 0.973]	0.00258	5	0
adaptive	seq2seq	0	0.971	[0.97, 0.972]	8.96×10^{-4}	5	0
adaptive	seq2seq	0.01	0.9737	[0.972, 0.975]	0.00144	5	0
adaptive	seq2seq	0.1	0.9736	[0.973, 0.975]	0.00121	5	0
adaptive	seq2seq	1	0.9738	[0.973, 0.975]	0.00132	5	0
adaptive	seq2seq	10	0.9736	[0.973, 0.975]	0.00145	5	0
adaptive	seq2seq	100	0.974	[0.973, 0.975]	0.00117	5	0
adaptive	seq2seq	1000	0.9742	[0.973, 0.975]	0.00123	5	0
adaptive	seq2seq	1e+04	0.9737	[0.972, 0.975]	0.00184	5	0
tropical	pointer	0	0.9713	[0.966, 0.977]	0.00697	5	0
tropical	pointer	0.01	0.9713	[0.967, 0.977]	0.00636	5	0
tropical	pointer	0.1	0.9688	[0.966, 0.971]	0.00299	5	0
tropical	pointer	1	0.971	[0.968, 0.975]	0.00412	5	0
tropical	pointer	10	0.9729	[0.967, 0.979]	0.00717	5	0
tropical	pointer	100	0.9711	[0.968, 0.975]	0.0043	5	0
tropical	pointer	1000	0.9726	[0.967, 0.978]	0.0067	5	0
tropical	pointer	1e+04	0.9709	[0.967, 0.977]	0.00676	5	0
tropical	seq2seq	0	0.9786	[0.973, 0.982]	0.00649	5	0
tropical	seq2seq	0.01	0.9783	[0.974, 0.982]	0.00513	5	0
tropical	seq2seq	0.1	0.978	[0.974, 0.982]	0.00519	5	0
tropical	seq2seq	1	0.9779	[0.973, 0.982]	0.00649	5	0
tropical	seq2seq	10	0.9763	[0.971, 0.982]	0.0076	5	0
tropical	seq2seq	100	0.9765	[0.972, 0.981]	0.00531	5	0
tropical	seq2seq	1000	0.9795	[0.977, 0.981]	0.00282	5	0
tropical	seq2seq	1e+04	0.9793	[0.975, 0.982]	0.005	5	0
vanilla	pointer	0	0.9702	[0.969, 0.971]	0.0013	5	0
vanilla	pointer	0.01	0.9714	[0.97, 0.973]	0.00236	5	0
vanilla	pointer	0.1	0.9713	[0.97, 0.973]	0.00222	5	0
vanilla	pointer	1	0.9715	[0.97, 0.973]	0.00203	5	0
vanilla	pointer	10	0.9712	[0.97, 0.973]	0.00224	5	0
vanilla	pointer	100	0.9713	[0.97, 0.973]	0.00213	5	0
vanilla	pointer	1000	0.9714	[0.97, 0.973]	0.00229	5	0
vanilla	pointer	1e+04	0.9714	[0.97, 0.973]	0.00219	5	0
vanilla	seq2seq	0	0.9725	[0.971, 0.974]	0.00168	5	0
vanilla	seq2seq	0.01	0.9747	[0.974, 0.976]	8.93×10^{-4}	5	0
vanilla	seq2seq	0.1	0.9746	[0.974, 0.975]	0.00101	5	0
vanilla	seq2seq	1	0.9748	[0.974, 0.976]	0.00131	5	0
vanilla	seq2seq	10	0.9749	[0.974, 0.976]	8.83×10^{-4}	5	0
vanilla	seq2seq	100	0.9747	[0.974, 0.975]	8.06×10^{-4}	5	0
vanilla	seq2seq	1000	0.9748	[0.974, 0.975]	7.16×10^{-4}	5	0
vanilla	seq2seq	1e+04	0.974	[0.973, 0.975]	0.00108	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	2.893×10^{-4}	[-0.00407, 0.00465]	1.0000
adaptive	pointer	0.1	3.051×10^{-4}	[-0.00405, 0.00466]	1.0000
adaptive	pointer	1	9.619×10^{-5}	[-0.00426, 0.00445]	1.0000
adaptive	pointer	10	3.542×10^{-4}	[-0.004, 0.00471]	1.0000
adaptive	pointer	100	1.845×10^{-4}	[-0.00417, 0.00454]	1.0000
adaptive	pointer	1000	9.402×10^{-5}	[-0.00426, 0.00445]	1.0000
adaptive	pointer	1e+04	2.668×10^{-4}	[-0.00409, 0.00463]	1.0000

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	seq2seq	0.01	0.002638	$[2.90 \times 10^{-4}, 0.00499]$	0.0223	*
adaptive	seq2seq	0.1	0.002575	$[2.28 \times 10^{-4}, 0.00492]$	0.0267	*
adaptive	seq2seq	1	0.002789	$[4.41 \times 10^{-4}, 0.00514]$	0.0143	*
adaptive	seq2seq	10	0.002616	$[2.69 \times 10^{-4}, 0.00496]$	0.0238	*
adaptive	seq2seq	100	0.003012	$[6.64 \times 10^{-4}, 0.00536]$	0.0073	*
adaptive	seq2seq	1000	0.003151	$[8.03 \times 10^{-4}, 0.0055]$	0.0048	*
adaptive	seq2seq	1e+04	0.002644	$[2.96 \times 10^{-4}, 0.00499]$	0.0221	*
tropical	pointer	0.01	-4.290×10^{-5}	[-0.0103, 0.0102]	1.0000	
tropical	pointer	0.1	-0.002482	[-0.0127, 0.00777]	0.9765	
tropical	pointer	1	-2.902×10^{-4}	[-0.0105, 0.00996]	1.0000	
tropical	pointer	10	0.001553	[-0.00869, 0.0118]	0.9984	
tropical	pointer	100	-2.028×10^{-4}	[-0.0105, 0.01]	1.0000	
tropical	pointer	1000	0.001254	[-0.00899, 0.0115]	0.9996	
tropical	pointer	1e+04	-4.138×10^{-4}	[-0.0107, 0.00983]	1.0000	
tropical	seq2seq	0.01	-2.560×10^{-4}	[-0.0101, 0.00963]	1.0000	
tropical	seq2seq	0.1	-5.735×10^{-4}	[-0.0105, 0.00932]	1.0000	
tropical	seq2seq	1	-7.054×10^{-4}	[-0.0106, 0.00919]	1.0000	
tropical	seq2seq	10	-0.002219	[-0.0121, 0.00767]	0.9843	
tropical	seq2seq	100	-0.002045	[-0.0119, 0.00785]	0.9900	
tropical	seq2seq	1000	9.785×10^{-4}	[-0.00891, 0.0109]	0.9999	
tropical	seq2seq	1e+04	7.532×10^{-4}	[-0.00914, 0.0106]	1.0000	
vanilla	pointer	0.01	0.001138	[-0.00256, 0.00484]	0.9257	
vanilla	pointer	0.1	0.001107	[-0.0026, 0.00481]	0.9345	
vanilla	pointer	1	0.001251	[-0.00245, 0.00495]	0.8890	
vanilla	pointer	10	9.675×10^{-4}	[-0.00273, 0.00467]	0.9654	
vanilla	pointer	100	0.001053	[-0.00265, 0.00476]	0.9478	
vanilla	pointer	1000	0.001212	[-0.00249, 0.00491]	0.9026	
vanilla	pointer	1e+04	0.001168	[-0.00253, 0.00487]	0.9166	
vanilla	seq2seq	0.01	0.002207	$[3.06 \times 10^{-4}, 0.00411]$	0.0173	*
vanilla	seq2seq	0.1	0.002095	$[1.95 \times 10^{-4}, 0.004]$	0.0257	*
vanilla	seq2seq	1	0.002276	$[3.76 \times 10^{-4}, 0.00418]$	0.0134	*
vanilla	seq2seq	10	0.002378	$[4.77 \times 10^{-4}, 0.00428]$	0.0092	*
vanilla	seq2seq	100	0.002184	$[2.83 \times 10^{-4}, 0.00408]$	0.0187	*
vanilla	seq2seq	1000	0.002245	$[3.44 \times 10^{-4}, 0.00415]$	0.0151	*
vanilla	seq2seq	1e+04	0.001513	$[-3.87 \times 10^{-4}, 0.00341]$	0.1641	

Table 11: `ood_val_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-1.433×10^{-5}	$[-4.67 \times 10^{-5}, 1.81 \times 10^{-5}]$	0.9432	-0.0171	0.9046
adaptive	seq2seq	4.971×10^{-5}	$[-3.10 \times 10^{-5}, 1.56 \times 10^{-4}]$	0.6538	0.0819	0.5607
tropical	pointer	2.302×10^{-4}	$[-2.54 \times 10^{-4}, 6.64 \times 10^{-4}]$	0.6154	0.0629	0.6513
tropical	seq2seq	1.712×10^{-4}	$[-1.15 \times 10^{-4}, 4.36 \times 10^{-4}]$	0.7005	0.0629	0.6507
vanilla	pointer	3.717×10^{-6}	$[-2.02 \times 10^{-5}, 2.80 \times 10^{-5}]$	0.9829	0.00571	0.9762
vanilla	seq2seq	-6.690×10^{-5}	$[-1.09 \times 10^{-4}, 5.95 \times 10^{-6}]$	0.4000	-0.0895	0.5063

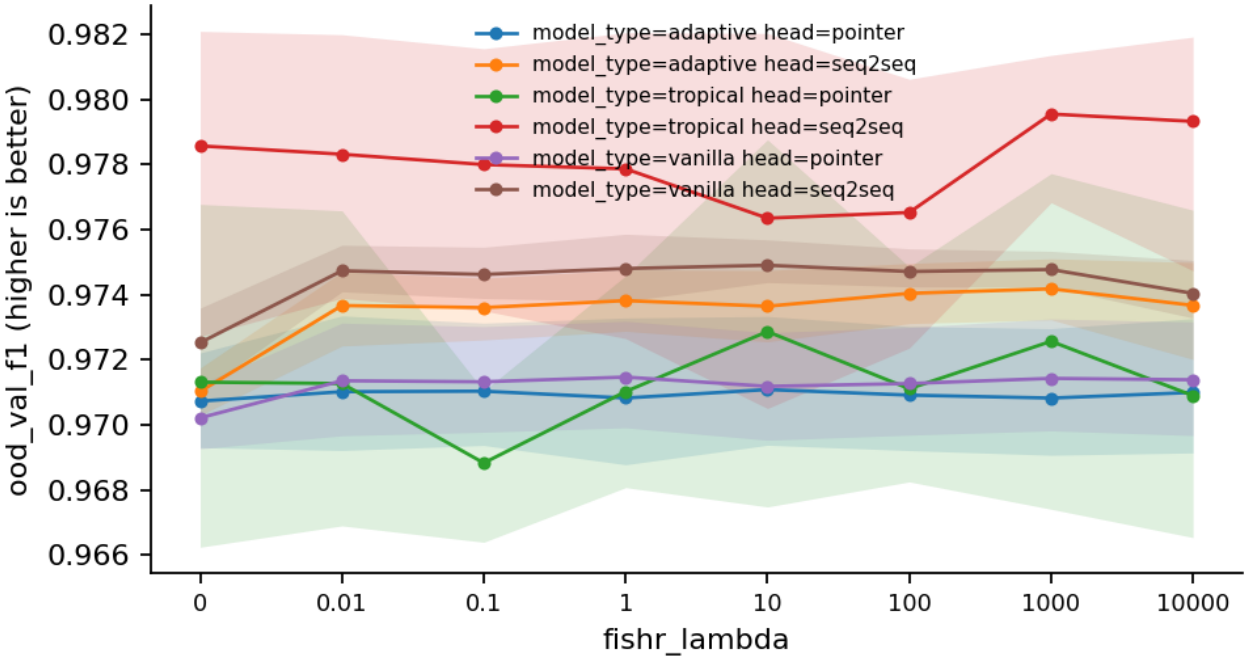


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.001096	[-0.008, 0.0102]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-8.468×10^{-5}	[-0.00534, 0.00517]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.002492	[-0.00485, -1.38×10^{-4}]	0.6367
model_type: tropical - vanilla head=pointer fishr_lambda=1	-4.450×10^{-4}	[-0.00456, 0.00367]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.001682	[-0.00683, 0.0102]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-1.597×10^{-4}	[-0.00511, 0.00479]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.001138	[-0.00581, 0.00809]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-4.861×10^{-4}	[-0.00643, 0.00545]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.006042	[-0.00234, 0.0144]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.003579	[-0.00215, 0.00931]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.003373	[-0.00337, 0.0101]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.00306	[-0.00379, 0.00991]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.001445	[-0.00737, 0.0103]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.001814	[-0.0042, 0.00783]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.004775	[0.00138, 0.00817]	0.2801
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.005282	[1.14×10^{-5}, 0.0106]	0.6957
model_type: adaptive - vanilla head=pointer fishr_lambda=0	5.132×10^{-4}	[-0.00227, 0.0033]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-3.355×10^{-4}	[-0.00198, 0.00131]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-2.882×10^{-4}	[-0.00159, 0.00101]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-6.416×10^{-4}	[-0.00261, 0.00133]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-1.000×10^{-4}	[-0.00132, 0.00112]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-3.554×10^{-4}	[-0.00144, 7.33×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-6.046×10^{-4}	[-0.00183, 6.25×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-3.884×10^{-4}	[-0.00227, 0.0015]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.001498	[-0.00404, 0.00104]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.001067	[-0.00208, -5.31×10^{-5}]	0.5612
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.001018	[-0.00154, -4.98×10^{-4}]	0.0889
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-9.855×10^{-4}	[-0.00179, -1.83×10^{-4}]	0.3781
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.00126	[-0.00226, -2.64×10^{-4}]	0.3691
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-6.698×10^{-4}	[-0.00184, 4.99×10^{-4}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-5.926×10^{-4}	[-0.00138, 1.95×10^{-4}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-3.676×10^{-4}	[-0.00292, 0.00218]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-2.973×10^{-4}	[-0.00257, 0.00197]	0.8163
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.002646	[-0.00597, 6.80×10^{-4}]	0.8163
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.002567	[-0.00569, 5.55×10^{-4}]	0.8163
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.00299	[-0.00641, 4.29×10^{-4}]	0.7942
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.002559	[-0.00617, 0.00105]	0.8163
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.003125	[-0.00555, -6.96×10^{-4}]	0.5142
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.003354	[-0.00634, -3.66×10^{-4}]	0.5345
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.002674	[-0.00713, 0.00178]	0.8163
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.007254	[-0.0222, 0.00771]	0.8163

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.007041	[-0.0156, 0.00156]	0.8163
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.009163	[-0.0162, -0.00214]	0.5142
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.006839	[-0.0144, 7.31×10^{-4}]	0.7942
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.003482	[-0.0103, 0.00329]	0.8163
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.005413	[-0.0129, 0.00207]	0.8163
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.006979	[-0.0134, -5.63×10^{-4}]	0.5483
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.008421	[-0.0169, 6.28×10^{-5}]	0.6638
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.002309	[-0.00416, -4.56×10^{-4}]	0.5152
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.003377	[-0.00614, -6.09×10^{-4}]	0.5152
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.003297	[-0.00594, -6.53×10^{-4}]	0.5152
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.003334	[-0.00609, -5.77×10^{-4}]	0.5152
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.003719	[-0.00652, -9.21×10^{-4}]	0.5044
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.003439	[-0.00639, -4.86×10^{-4}]	0.5152
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.003342	[-0.00596, -7.28×10^{-4}]	0.5142
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.002653	[-0.00584, 5.30×10^{-4}]	0.8163

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8848	[0.872, 0.896]	0.0154	5	0
adaptive	pointer	0.01	0.8852	[0.872, 0.9]	0.0187	5	0
adaptive	pointer	0.1	0.8856	[0.873, 0.899]	0.0174	5	0
adaptive	pointer	1	0.8851	[0.871, 0.9]	0.0203	5	0
adaptive	pointer	10	0.886	[0.873, 0.901]	0.0172	5	0
adaptive	pointer	100	0.8849	[0.872, 0.898]	0.0173	5	0
adaptive	pointer	1000	0.8839	[0.872, 0.897]	0.0163	5	0
adaptive	pointer	1e+04	0.8861	[0.873, 0.9]	0.0173	5	0
adaptive	seq2seq	0	0.8811	[0.87, 0.892]	0.0148	5	0
adaptive	seq2seq	0.01	0.8991	[0.89, 0.908]	0.0119	5	0
adaptive	seq2seq	0.1	0.8993	[0.89, 0.907]	0.0115	5	0
adaptive	seq2seq	1	0.9015	[0.891, 0.91]	0.0116	5	0
adaptive	seq2seq	10	0.8998	[0.891, 0.909]	0.0116	5	0
adaptive	seq2seq	100	0.9028	[0.894, 0.91]	0.0112	5	0
adaptive	seq2seq	1000	0.9032	[0.894, 0.911]	0.0105	5	0
adaptive	seq2seq	1e+04	0.896	[0.888, 0.904]	0.0107	5	0
tropical	pointer	0	0.9029	[0.883, 0.929]	0.029	5	0
tropical	pointer	0.01	0.9002	[0.877, 0.932]	0.0366	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.8873	[0.878, 0.894]	0.0102	5	0
tropical	pointer	1	0.899	[0.88, 0.92]	0.0256	5	0
tropical	pointer	10	0.9086	[0.872, 0.947]	0.048	5	0
tropical	pointer	100	0.8884	[0.865, 0.918]	0.0323	5	0
tropical	pointer	1000	0.8982	[0.866, 0.928]	0.042	5	0
tropical	pointer	1e+04	0.9026	[0.877, 0.938]	0.0402	5	0
tropical	seq2seq	0	0.9489	[0.915, 0.97]	0.0369	5	0
tropical	seq2seq	0.01	0.9405	[0.903, 0.969]	0.0418	5	0
tropical	seq2seq	0.1	0.9355	[0.896, 0.968]	0.0462	5	0
tropical	seq2seq	1	0.9444	[0.913, 0.969]	0.0367	5	0
tropical	seq2seq	10	0.9351	[0.901, 0.969]	0.0458	5	0
tropical	seq2seq	100	0.9244	[0.892, 0.954]	0.0407	5	0
tropical	seq2seq	1000	0.9437	[0.911, 0.965]	0.0364	5	0
tropical	seq2seq	1e+04	0.9475	[0.909, 0.969]	0.042	5	0
vanilla	pointer	0	0.8784	[0.872, 0.884]	0.00747	5	0
vanilla	pointer	0.01	0.8876	[0.877, 0.9]	0.0154	5	0
vanilla	pointer	0.1	0.8855	[0.874, 0.897]	0.0156	5	0
vanilla	pointer	1	0.8877	[0.878, 0.899]	0.0137	5	0
vanilla	pointer	10	0.8852	[0.873, 0.898]	0.0168	5	0
vanilla	pointer	100	0.8864	[0.874, 0.9]	0.0166	5	0
vanilla	pointer	1000	0.8874	[0.875, 0.9]	0.0167	5	0
vanilla	pointer	1e+04	0.8887	[0.877, 0.902]	0.0158	5	0
vanilla	seq2seq	0	0.8943	[0.887, 0.902]	0.00983	5	0
vanilla	seq2seq	0.01	0.9105	[0.903, 0.917]	0.00883	5	0
vanilla	seq2seq	0.1	0.9099	[0.902, 0.917]	0.00928	5	0
vanilla	seq2seq	1	0.91	[0.898, 0.92]	0.0147	5	0
vanilla	seq2seq	10	0.9107	[0.902, 0.918]	0.0105	5	0
vanilla	seq2seq	100	0.91	[0.904, 0.915]	0.00784	5	0
vanilla	seq2seq	1000	0.9098	[0.905, 0.914]	0.00538	5	0
vanilla	seq2seq	1e+04	0.903	[0.892, 0.912]	0.0138	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	4.400×10^{-4}	[-0.0302, 0.0311]	1.0000	
adaptive	pointer	0.1	8.600×10^{-4}	[-0.0298, 0.0315]	1.0000	
adaptive	pointer	1	3.000×10^{-4}	[-0.0303, 0.0309]	1.0000	
adaptive	pointer	10	0.00122	[-0.0294, 0.0319]	1.0000	
adaptive	pointer	100	1.000×10^{-4}	[-0.0305, 0.0307]	1.0000	
adaptive	pointer	1000	-8.400×10^{-4}	[-0.0315, 0.0298]	1.0000	
adaptive	pointer	1e+04	0.00134	[-0.0293, 0.032]	1.0000	
adaptive	seq2seq	0.01	0.018	[-0.00265, 0.0386]	0.1081	
adaptive	seq2seq	0.1	0.01822	[-0.00243, 0.0389]	0.1015	
adaptive	seq2seq	1	0.0204	$[-2.48 \times 10^{-4}, 0.041]$	0.0539	
adaptive	seq2seq	10	0.0187	[-0.00195, 0.0393]	0.0889	
adaptive	seq2seq	100	0.02172	[0.00107, 0.0424]	0.0358	*
adaptive	seq2seq	1000	0.02214	[0.00149, 0.0428]	0.0314	*
adaptive	seq2seq	1e+04	0.01488	[-0.00577, 0.0355]	0.2414	
tropical	pointer	0.01	-0.00272	[-0.0634, 0.058]	1.0000	
tropical	pointer	0.1	-0.01562	[-0.0763, 0.0451]	0.9680	
tropical	pointer	1	-0.00398	[-0.0647, 0.0567]	1.0000	
tropical	pointer	10	0.00562	[-0.0551, 0.0663]	0.9999	
tropical	pointer	100	-0.01456	[-0.0753, 0.0462]	0.9777	

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	pointer	1000	-0.00478	[-0.0655, 0.0559]	1.0000
tropical	pointer	1e+04	-3.000×10^{-4}	[-0.061, 0.0604]	1.0000
tropical	seq2seq	0.01	-0.00842	[-0.08, 0.0632]	0.9997
tropical	seq2seq	0.1	-0.01342	[-0.085, 0.0582]	0.9943
tropical	seq2seq	1	-0.00458	[-0.0762, 0.067]	1.0000
tropical	seq2seq	10	-0.01386	[-0.0854, 0.0577]	0.9931
tropical	seq2seq	100	-0.02458	[-0.0962, 0.047]	0.8815
tropical	seq2seq	1000	-0.0052	[-0.0768, 0.0664]	1.0000
tropical	seq2seq	1e+04	-0.0014	[-0.073, 0.0702]	1.0000
vanilla	pointer	0.01	0.00914	[-0.0171, 0.0354]	0.8753
vanilla	pointer	0.1	0.00702	[-0.0193, 0.0333]	0.9614
vanilla	pointer	1	0.0093	[-0.017, 0.0356]	0.8666
vanilla	pointer	10	0.00678	[-0.0195, 0.0331]	0.9675
vanilla	pointer	100	0.00798	[-0.0183, 0.0343]	0.9296
vanilla	pointer	1000	0.00896	[-0.0173, 0.0352]	0.8848
vanilla	pointer	1e+04	0.01024	[-0.016, 0.0365]	0.8098
vanilla	seq2seq	0.01	0.0162	[-0.00196, 0.0344]	0.0960
vanilla	seq2seq	0.1	0.01566	[-0.0025, 0.0338]	0.1138
vanilla	seq2seq	1	0.01576	[-0.0024, 0.0339]	0.1104
vanilla	seq2seq	10	0.01642	[-0.00174, 0.0346]	0.0895
vanilla	seq2seq	100	0.01576	[-0.0024, 0.0339]	0.1105
vanilla	seq2seq	1000	0.0155	[-0.00266, 0.0337]	0.1197
vanilla	seq2seq	1e+04	0.0087	[-0.00946, 0.0269]	0.6422

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-3.214×10^{-5}	$[-2.82 \times 10^{-4}, 2.11 \times 10^{-4}]$	0.9817	0.00762	0.9634
adaptive	seq2seq	-7.143×10^{-6}	$[-6.06 \times 10^{-4}, 8.13 \times 10^{-4}]$	0.9937	0.0152	0.9182
tropical	pointer	6.557×10^{-4}	[-0.00179, 0.00265]	0.8181	0.0457	0.7419
tropical	seq2seq	6.250×10^{-4}	[-0.00205, 0.00291]	0.8506	0.04	0.7690
vanilla	pointer	2.093×10^{-4}	$[-5.57 \times 10^{-5}, 4.35 \times 10^{-4}]$	0.8666	0.0381	0.7806
vanilla	seq2seq	-8.150×10^{-4}	$[-0.00118, -4.05 \times 10^{-4}]$	0.3347	-0.107	0.4189

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0245	[-0.0168, 0.0658]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.01264	[-0.0176, 0.0428]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.00186	[-0.0129, 0.0166]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.01122	[-0.0128, 0.0353]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.02334	[-0.0331, 0.0798]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.00196	[-0.03, 0.034]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.01076	[-0.0315, 0.0531]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.01396	[-0.0208, 0.0487]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.05466	[0.0179, 0.0914]	0.2327
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.03004	[-0.0132, 0.0733]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.02558	[-0.0284, 0.0796]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.03432	[0.00284, 0.0658]	0.5448
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.02438	[-0.0262, 0.075]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.01432	[-0.0305, 0.0591]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.03396	[-0.00548, 0.0734]	0.9762
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.04456	[0.0083, 0.0808]	0.4047
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00634	[-0.00733, 0.02]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.00236	[-0.0124, 0.00765]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	1.800×10^{-4}	[-0.00626, 0.00662]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.00266	[-0.0126, 0.00727]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	7.800×10^{-4}	[-0.00538, 0.00694]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.00154	[-0.00881, 0.00573]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.00346	[-0.00884, 0.00192]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.00256	[-0.0113, 0.00622]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.01318	[-0.0337, 0.00732]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.01138	[-0.0207, -0.0021]	0.4077
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01062	[-0.0174, -0.00388]	0.1908
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.00854	[-0.0168, -2.64×10^{-4}]	0.5941
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.0109	[-0.0202, -0.00156]	0.4439
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00722	[-0.0157, 0.00127]	0.9316
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00654	[-0.0161, 0.003]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.007	[-0.0261, 0.0121]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.00368	[-0.0155, 0.0229]	0.9298
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.01388	[-0.0358, 0.00803]	0.9298

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.01368	[-0.0337, 0.00637]	0.9298
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.01642	[-0.0373, 0.00444]	0.9298
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0138	[-0.0367, 0.00912]	0.9298
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.01794	[-0.0323, -0.00355]	0.5296
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.0193	[-0.0364, -0.00218]	0.6329
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.00986	[-0.0371, 0.0174]	0.9298
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.046	[-0.117, 0.0249]	0.9298
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.0403	[-0.0826, 0.00199]	0.6866
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.0482	[-0.107, 0.0104]	0.9298
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.0454	[-0.0748, -0.016]	0.2814
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.02652	[-0.0631, 0.0101]	0.9298
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.03598	[-0.0697, -0.00227]	0.6329
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.04558	[-0.0706, -0.0205]	0.1738
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.0449	[-0.1, 0.0102]	0.9298
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.01584	[-0.0253, -0.00634]	0.2260
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.0229	[-0.044, -0.00177]	0.6329
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.02448	[-0.044, -0.00498]	0.5296
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.0223	$[-0.0446, 8.87 \times 10^{-6}]$	0.6507
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.02548	[-0.0466, -0.0044]	0.5394
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.02362	[-0.0446, -0.00262]	0.6329
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.02238	[-0.0424, -0.00241]	0.6329
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.0143	[-0.0396, 0.011]	0.9298

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.514	[0.469, 0.559]	0.0594	5	0
adaptive	pointer	0.01	0.4725	[0.288, 0.65]	0.235	5	0
adaptive	pointer	0.1	0.4715	[0.29, 0.629]	0.22	5	0
adaptive	pointer	1	0.4577	[0.286, 0.638]	0.216	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.4831	[0.293, 0.645]	0.216	5	0
adaptive	pointer	100	0.4907	[0.298, 0.641]	0.217	5	0
adaptive	pointer	1000	0.5465	[0.359, 0.693]	0.223	5	0
adaptive	pointer	1e+04	0.422	[0.209, 0.639]	0.283	5	0
adaptive	seq2seq	0	0.5588	[0.44, 0.685]	0.156	5	0
adaptive	seq2seq	0.01	0.566	[0.511, 0.621]	0.0702	5	0
adaptive	seq2seq	0.1	0.5444	[0.507, 0.581]	0.0455	5	0
adaptive	seq2seq	1	0.5509	[0.488, 0.585]	0.0694	5	0
adaptive	seq2seq	10	0.5637	[0.498, 0.623]	0.0872	5	0
adaptive	seq2seq	100	0.5551	[0.503, 0.61]	0.0709	5	0
adaptive	seq2seq	1000	0.5448	[0.484, 0.616]	0.088	5	0
adaptive	seq2seq	1e+04	0.481	[0.442, 0.507]	0.0432	5	0
tropical	pointer	0	0.591	[0.392, 0.749]	0.239	5	0
tropical	pointer	0.01	0.609	[0.332, 0.886]	0.366	5	0
tropical	pointer	0.1	0.7719	[0.581, 0.963]	0.244	5	0
tropical	pointer	1	0.6346	[0.376, 0.876]	0.327	5	0
tropical	pointer	10	0.681	[0.586, 0.767]	0.115	5	0
tropical	pointer	100	0.6142	[0.401, 0.827]	0.282	5	0
tropical	pointer	1000	0.6355	[0.469, 0.802]	0.217	5	0
tropical	pointer	1e+04	0.6705	[0.576, 0.775]	0.129	5	0
tropical	seq2seq	0	0.6521	[0.566, 0.807]	0.174	5	0
tropical	seq2seq	0.01	0.6565	[0.572, 0.796]	0.155	5	0
tropical	seq2seq	0.1	0.6585	[0.584, 0.782]	0.139	5	0
tropical	seq2seq	1	0.684	[0.572, 0.83]	0.183	5	0
tropical	seq2seq	10	0.736	[0.57, 0.905]	0.22	5	0
tropical	seq2seq	100	0.7144	[0.6, 0.843]	0.165	5	0
tropical	seq2seq	1000	0.6339	[0.6, 0.676]	0.0458	5	0
tropical	seq2seq	1e+04	0.6453	[0.558, 0.78]	0.146	5	0
vanilla	pointer	0	0.4555	[0.3, 0.602]	0.196	5	0
vanilla	pointer	0.01	0.4968	[0.342, 0.651]	0.198	5	0
vanilla	pointer	0.1	0.5012	[0.337, 0.659]	0.209	5	0
vanilla	pointer	1	0.489	[0.336, 0.642]	0.198	5	0
vanilla	pointer	10	0.4223	[0.245, 0.604]	0.218	5	0
vanilla	pointer	100	0.4865	[0.345, 0.639]	0.192	5	0
vanilla	pointer	1000	0.4653	[0.313, 0.638]	0.213	5	0
vanilla	pointer	1e+04	0.4806	[0.284, 0.677]	0.25	5	0
vanilla	seq2seq	0	0.5075	[0.451, 0.575]	0.0823	5	0
vanilla	seq2seq	0.01	0.5992	[0.516, 0.682]	0.113	5	0
vanilla	seq2seq	0.1	0.5837	[0.519, 0.647]	0.0894	5	0
vanilla	seq2seq	1	0.5601	[0.493, 0.635]	0.0883	5	0
vanilla	seq2seq	10	0.5479	[0.474, 0.624]	0.0925	5	0
vanilla	seq2seq	100	0.568	[0.498, 0.617]	0.0795	5	0
vanilla	seq2seq	1000	0.6303	[0.583, 0.679]	0.0657	5	0
vanilla	seq2seq	1e+04	0.5945	[0.531, 0.662]	0.0867	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.04148	[-0.421, 0.338]	0.9998
adaptive	pointer	0.1	-0.04248	[-0.422, 0.337]	0.9998
adaptive	pointer	1	-0.05634	[-0.436, 0.323]	0.9986
adaptive	pointer	10	-0.03088	[-0.41, 0.348]	1.0000
adaptive	pointer	100	-0.02337	[-0.403, 0.356]	1.0000

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	1000	0.03244	[-0.347, 0.412]	1.0000
adaptive	pointer	1e+04	-0.09205	[-0.471, 0.287]	0.9762
adaptive	seq2seq	0.01	0.00718	[-0.142, 0.156]	1.0000
adaptive	seq2seq	0.1	-0.01441	[-0.164, 0.135]	0.9999
adaptive	seq2seq	1	-0.007882	[-0.157, 0.141]	1.0000
adaptive	seq2seq	10	0.004937	[-0.144, 0.154]	1.0000
adaptive	seq2seq	100	-0.003685	[-0.153, 0.146]	1.0000
adaptive	seq2seq	1000	-0.01398	[-0.163, 0.135]	0.9999
adaptive	seq2seq	1e+04	-0.07775	[-0.227, 0.0715]	0.5607
tropical	pointer	0.01	0.01799	[-0.425, 0.461]	1.0000
tropical	pointer	0.1	0.1809	[-0.262, 0.624]	0.7767
tropical	pointer	1	0.04363	[-0.399, 0.486]	0.9999
tropical	pointer	10	0.09004	[-0.353, 0.533]	0.9909
tropical	pointer	100	0.0232	[-0.42, 0.466]	1.0000
tropical	pointer	1000	0.04455	[-0.398, 0.487]	0.9999
tropical	pointer	1e+04	0.07948	[-0.363, 0.522]	0.9956
tropical	seq2seq	0.01	0.004421	[-0.276, 0.285]	1.0000
tropical	seq2seq	0.1	0.006455	[-0.274, 0.287]	1.0000
tropical	seq2seq	1	0.03195	[-0.248, 0.312]	0.9997
tropical	seq2seq	10	0.08392	[-0.196, 0.364]	0.9339
tropical	seq2seq	100	0.06238	[-0.218, 0.343]	0.9850
tropical	seq2seq	1000	-0.01818	[-0.298, 0.262]	1.0000
tropical	seq2seq	1e+04	-0.006761	[-0.287, 0.273]	1.0000
vanilla	pointer	0.01	0.04127	[-0.326, 0.408]	0.9998
vanilla	pointer	0.1	0.04576	[-0.321, 0.413]	0.9995
vanilla	pointer	1	0.03348	[-0.333, 0.4]	0.9999
vanilla	pointer	10	-0.03316	[-0.4, 0.334]	0.9999
vanilla	pointer	100	0.03104	[-0.336, 0.398]	1.0000
vanilla	pointer	1000	0.009812	[-0.357, 0.377]	1.0000
vanilla	pointer	1e+04	0.02515	[-0.342, 0.392]	1.0000
vanilla	seq2seq	0.01	0.0917	[-0.0622, 0.246]	0.4226
vanilla	seq2seq	0.1	0.07618	[-0.0777, 0.23]	0.6109
vanilla	seq2seq	1	0.05263	[-0.101, 0.206]	0.8834
vanilla	seq2seq	10	0.04038	[-0.113, 0.194]	0.9646
vanilla	seq2seq	100	0.0605	[-0.0934, 0.214]	0.8037
vanilla	seq2seq	1000	0.1227	[-0.0311, 0.277]	0.1627
vanilla	seq2seq	1e+04	0.087	[-0.0669, 0.241]	0.4767

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.001111	[-0.0117, 0.0125]	0.9518	0.0895	0.5175
adaptive	seq2seq	-0.00892	[-0.0194, 0.00125]	0.1271	-0.204	0.1290
tropical	pointer	-0.003882	[-0.043, 0.0385]	0.8502	-0.0705	0.6019
tropical	seq2seq	-0.001871	[-0.011, 0.00384]	0.8838	0.0514	0.7161
vanilla	pointer	-0.004381	[-0.0165, 0.00469]	0.7938	-0.059	0.6565
vanilla	seq2seq	0.003103	[-0.0142, 0.0225]	0.6714	0.0552	0.6961

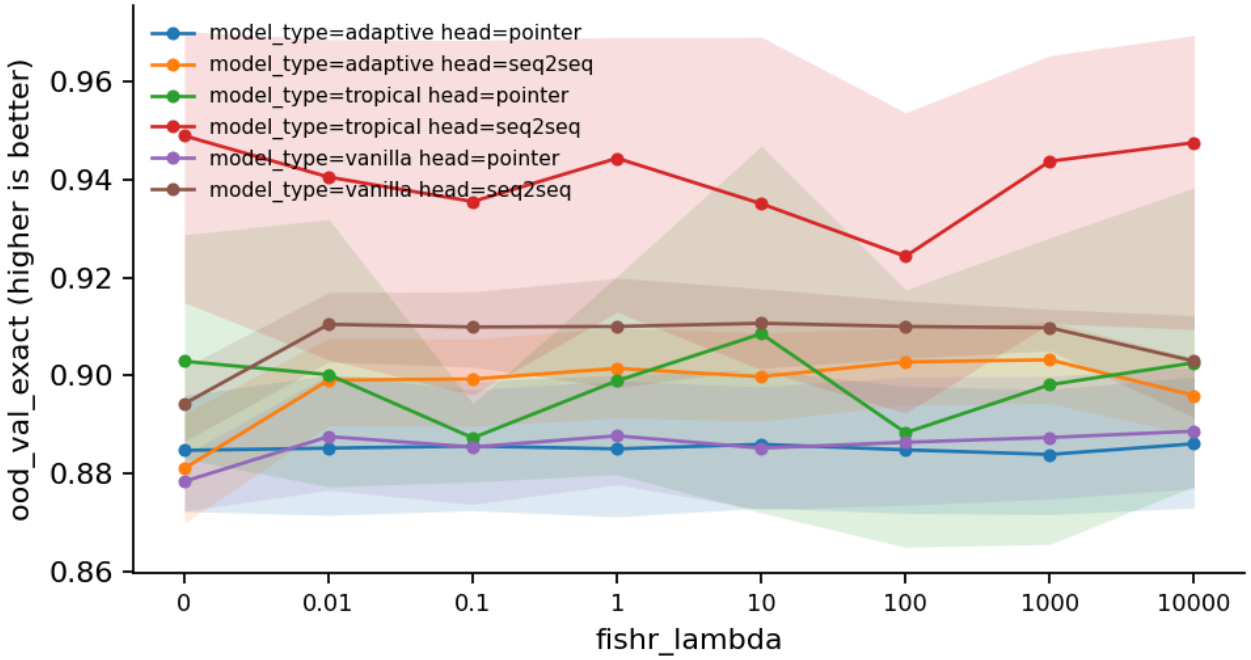


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

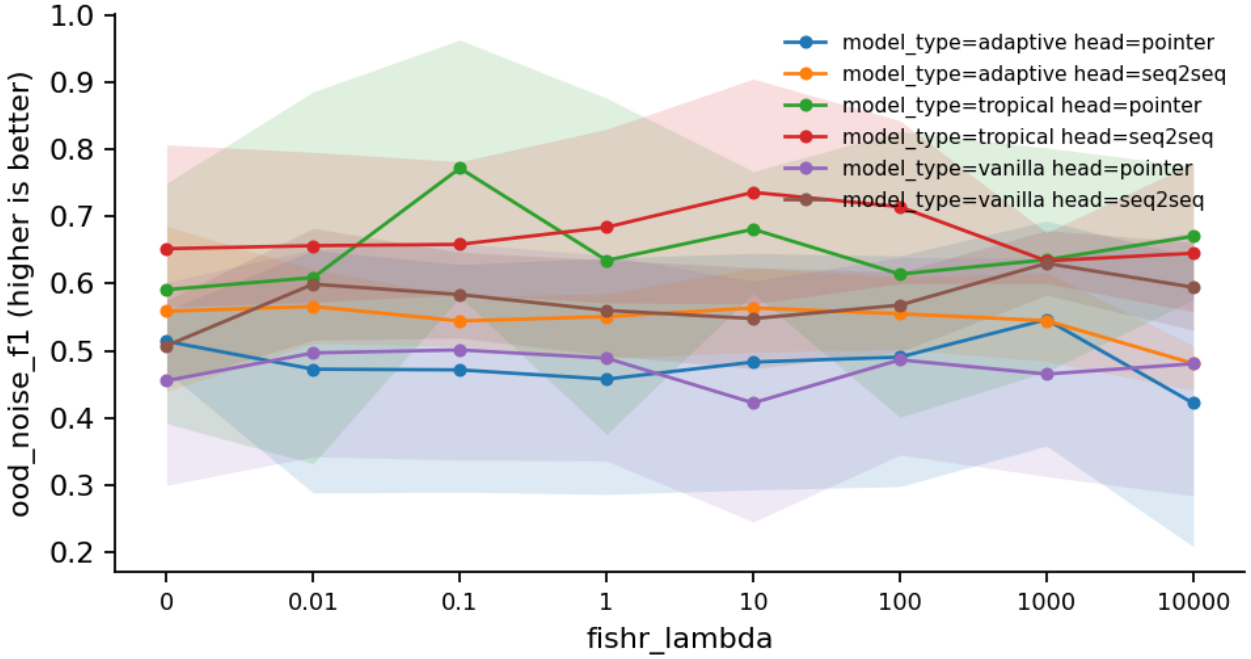


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.1355	[-0.303, 0.574]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.1122	[-0.353, 0.577]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.2707	[-0.0843, 0.626]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1457	[-0.246, 0.538]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.2587	[-0.0612, 0.579]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1277	[-0.382, 0.637]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.1702	[-0.0904, 0.431]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.1898	[-0.202, 0.582]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1445	[-0.11, 0.4]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.05726	[-0.0888, 0.203]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.07482	[-0.115, 0.264]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1239	[-0.107, 0.354]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1881	[-0.0402, 0.416]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1464	[-0.00915, 0.302]	0.9476
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.003626	[-0.0557, 0.063]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.05078	[-0.203, 0.304]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.05853	[-0.217, 0.334]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.02422	[-0.305, 0.257]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.02971	[-0.372, 0.313]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.0313	[-0.297, 0.234]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.06081	[-0.248, 0.37]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.004116	[-0.243, 0.252]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.08116	[-0.199, 0.361]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.05868	[-0.544, 0.427]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.05128	[-0.178, 0.281]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.03324	[-0.241, 0.175]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.03931	[-0.183, 0.105]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.009235	[-0.15, 0.131]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01584	[-0.161, 0.193]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0129	[-0.153, 0.128]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.08544	[-0.26, 0.0888]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.1135	[-0.26, 0.0329]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.04477	[-0.228, 0.138]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.09344	[-0.435, 0.248]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.07285	[-0.337, 0.192]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.09323	[-0.383, 0.196]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.08059	[-0.395, 0.234]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.06446	[-0.364, 0.235]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.00165	[-0.237, 0.24]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.05907	[-0.395, 0.277]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.06106	[-0.359, 0.237]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.04749	[-0.585, 0.49]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.1134	[-0.129, 0.356]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.04938	[-0.639, 0.54]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.05494	[-0.298, 0.189]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.1002	[-0.651, 0.451]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.001671	[-0.301, 0.304]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.02518	[-0.166, 0.217]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.05202	[-0.352, 0.248]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.1025	[-0.335, 0.13]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.08244	[-0.303, 0.138]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.07117	[-0.26, 0.118]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.1256	[-0.355, 0.104]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.08148	[-0.276, 0.113]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.1649	[-0.469, 0.139]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.1139	[-0.475, 0.247]	1.0000

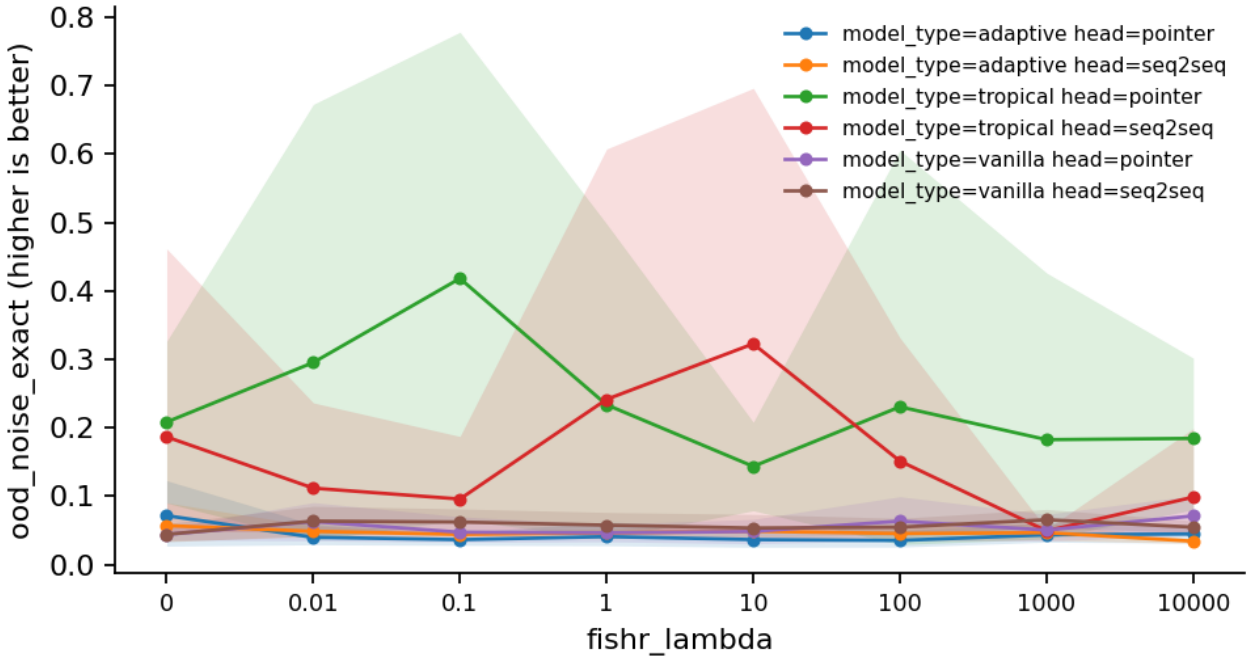


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.07172	[0.0273, 0.123]	0.0607	5	0
adaptive	pointer	0.01	0.04008	[0.0294, 0.0589]	0.0207	5	0
adaptive	pointer	0.1	0.03662	[0.0285, 0.0465]	0.0115	5	0
adaptive	pointer	1	0.04104	[0.0279, 0.0629]	0.0231	5	0
adaptive	pointer	10	0.0367	[0.0257, 0.0533]	0.0188	5	0
adaptive	pointer	100	0.03546	[0.0259, 0.0501]	0.0166	5	0
adaptive	pointer	1000	0.04368	[0.0331, 0.0613]	0.0198	5	0
adaptive	pointer	1e+04	0.04498	[0.0317, 0.0582]	0.0174	5	0
adaptive	seq2seq	0	0.05714	[0.0345, 0.0906]	0.0376	5	0
adaptive	seq2seq	0.01	0.0488	[0.041, 0.0566]	0.00996	5	0
adaptive	seq2seq	0.1	0.04422	[0.0398, 0.0488]	0.00593	5	0
adaptive	seq2seq	1	0.0465	[0.038, 0.0526]	0.00963	5	0
adaptive	seq2seq	10	0.04894	[0.0409, 0.057]	0.0111	5	0
adaptive	seq2seq	100	0.04556	[0.0359, 0.0553]	0.0128	5	0
adaptive	seq2seq	1000	0.04684	[0.036, 0.0606]	0.0166	5	0
adaptive	seq2seq	1e+04	0.03438	[0.0311, 0.0376]	0.00409	5	0
tropical	pointer	0	0.2085	[0.0904, 0.327]	0.152	5	0
tropical	pointer	0.01	0.2955	[0.0356, 0.673]	0.412	5	0
tropical	pointer	0.1	0.4184	[0.0655, 0.778]	0.478	5	0
tropical	pointer	1	0.2338	[0.0383, 0.498]	0.297	5	0
tropical	pointer	10	0.1437	[0.0791, 0.208]	0.0851	5	0
tropical	pointer	100	0.2308	[0.0296, 0.606]	0.417	5	0
tropical	pointer	1000	0.1828	[0.038, 0.426]	0.267	5	0
tropical	pointer	1e+04	0.1846	[0.0805, 0.302]	0.15	5	0
tropical	seq2seq	0	0.1868	[0.0488, 0.462]	0.307	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.1119	[0.0478, 0.237]	0.139	5	0
tropical	seq2seq	0.1	0.096	[0.0491, 0.188]	0.102	5	0
tropical	seq2seq	1	0.2417	[0.0487, 0.608]	0.412	5	0
tropical	seq2seq	10	0.3228	[0.049, 0.696]	0.414	5	0
tropical	seq2seq	100	0.1515	[0.0462, 0.332]	0.196	5	0
tropical	seq2seq	1000	0.04938	[0.0466, 0.0521]	0.00348	5	0
tropical	seq2seq	1e+04	0.09878	[0.0467, 0.199]	0.111	5	0
vanilla	pointer	0	0.04424	[0.0363, 0.0538]	0.0111	5	0
vanilla	pointer	0.01	0.06304	[0.0351, 0.0909]	0.0364	5	0
vanilla	pointer	0.1	0.04746	[0.033, 0.0702]	0.0255	5	0
vanilla	pointer	1	0.0466	[0.0337, 0.0595]	0.0172	5	0
vanilla	pointer	10	0.04856	[0.0307, 0.0664]	0.0227	5	0
vanilla	pointer	100	0.06372	[0.0353, 0.0994]	0.0423	5	0
vanilla	pointer	1000	0.05062	[0.0348, 0.0738]	0.0257	5	0
vanilla	pointer	1e+04	0.07126	[0.0433, 0.101]	0.0375	5	0
vanilla	seq2seq	0	0.04438	[0.0337, 0.0602]	0.0181	5	0
vanilla	seq2seq	0.01	0.06372	[0.0424, 0.0855]	0.0285	5	0
vanilla	seq2seq	0.1	0.06236	[0.0471, 0.0817]	0.0226	5	0
vanilla	seq2seq	1	0.05784	[0.0447, 0.0766]	0.0215	5	0
vanilla	seq2seq	10	0.0536	[0.0398, 0.0737]	0.023	5	0
vanilla	seq2seq	100	0.0549	[0.0434, 0.0678]	0.0171	5	0
vanilla	seq2seq	1000	0.06574	[0.0549, 0.081]	0.0176	5	0
vanilla	seq2seq	1e+04	0.05486	[0.0468, 0.0643]	0.0115	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.03164	[-0.0799, 0.0166]	0.3278
adaptive	pointer	0.1	-0.0351	[-0.0834, 0.0132]	0.2337
adaptive	pointer	1	-0.03068	[-0.0789, 0.0176]	0.3578
adaptive	pointer	10	-0.03502	[-0.0833, 0.0132]	0.2358
adaptive	pointer	100	-0.03626	[-0.0845, 0.012]	0.2074
adaptive	pointer	1000	-0.02804	[-0.0763, 0.0202]	0.4489
adaptive	pointer	1e+04	-0.02674	[-0.075, 0.0215]	0.4978
adaptive	seq2seq	0.01	-0.00834	[-0.0375, 0.0208]	0.9462
adaptive	seq2seq	0.1	-0.01292	[-0.042, 0.0162]	0.7116
adaptive	seq2seq	1	-0.01064	[-0.0398, 0.0185]	0.8495
adaptive	seq2seq	10	-0.0082	[-0.0373, 0.0209]	0.9503
adaptive	seq2seq	100	-0.01158	[-0.0407, 0.0175]	0.7962
adaptive	seq2seq	1000	-0.0103	[-0.0394, 0.0188]	0.8671
adaptive	seq2seq	1e+04	-0.02276	[-0.0519, 0.00637]	0.1774
tropical	pointer	0.01	0.087	[-0.46, 0.634]	0.9979
tropical	pointer	0.1	0.2099	[-0.337, 0.757]	0.8200
tropical	pointer	1	0.02528	[-0.522, 0.572]	1.0000
tropical	pointer	10	-0.06476	[-0.612, 0.482]	0.9997
tropical	pointer	100	0.0223	[-0.525, 0.569]	1.0000
tropical	pointer	1000	-0.02572	[-0.573, 0.521]	1.0000
tropical	pointer	1e+04	-0.02388	[-0.571, 0.523]	1.0000
tropical	seq2seq	0.01	-0.07492	[-0.519, 0.369]	0.9969
tropical	seq2seq	0.1	-0.09084	[-0.535, 0.353]	0.9906
tropical	seq2seq	1	0.05482	[-0.389, 0.499]	0.9996
tropical	seq2seq	10	0.136	[-0.308, 0.58]	0.9268
tropical	seq2seq	100	-0.03534	[-0.479, 0.409]	1.0000

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	seq2seq	1000	-0.1375	[-0.581, 0.306]	0.9233
tropical	seq2seq	1e+04	-0.08806	[-0.532, 0.356]	0.9921
vanilla	pointer	0.01	0.0188	[-0.032, 0.0696]	0.8421
vanilla	pointer	0.1	0.00322	[-0.0476, 0.054]	1.0000
vanilla	pointer	1	0.00236	[-0.0485, 0.0532]	1.0000
vanilla	pointer	10	0.00432	[-0.0465, 0.0551]	1.0000
vanilla	pointer	100	0.01948	[-0.0313, 0.0703]	0.8204
vanilla	pointer	1000	0.00638	[-0.0444, 0.0572]	0.9995
vanilla	pointer	1e+04	0.02702	[-0.0238, 0.0778]	0.5397
vanilla	seq2seq	0.01	0.01934	[-0.0165, 0.0552]	0.5261
vanilla	seq2seq	0.1	0.01798	[-0.0179, 0.0539]	0.5994
vanilla	seq2seq	1	0.01346	[-0.0224, 0.0493]	0.8341
vanilla	seq2seq	10	0.00922	[-0.0267, 0.0451]	0.9682
vanilla	seq2seq	100	0.01052	[-0.0254, 0.0464]	0.9400
vanilla	seq2seq	1000	0.02136	[-0.0145, 0.0572]	0.4241
vanilla	seq2seq	1e+04	0.01048	[-0.0254, 0.0464]	0.9411

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	8.300×10^{-4}	$[-1.45 \times 10^{-4}, 0.0023]$	0.5765	0.101	0.4583
adaptive	seq2seq	-0.001391	$[-0.00276, -4.24 \times 10^{-4}]$	0.1298	-0.234	0.0840
tropical	pointer	-0.02881	[-0.075, 0.00891]	0.2802	-0.0324	0.8198
tropical	seq2seq	-0.007958	$[-0.0244, 6.32 \times 10^{-4}]$	0.7015	-0.0552	0.6899
vanilla	pointer	0.001718	[-0.00167, 0.00511]	0.4985	0.0781	0.5717
vanilla	seq2seq	-8.129×10^{-4}	[-0.00458, 0.00278]	0.6289	0.0171	0.9078

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.1642	[-0.0185, 0.347]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.2324	[-0.299, 0.764]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.3709	[-0.239, 0.981]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1872	[-0.19, 0.564]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.09516	[0.00953, 0.181]	0.5878
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1671	[-0.378, 0.712]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.1321	[-0.169, 0.433]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.1133	[-0.104, 0.331]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1425	[-0.249, 0.534]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.0482	[-0.108, 0.204]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.03364	[-0.0939, 0.161]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1838	[-0.334, 0.702]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.2692	[-0.241, 0.779]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.0966	[-0.141, 0.334]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.01636	[-0.0402, 0.00753]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.04392	[-0.101, 0.189]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.02748	[-0.0528, 0.108]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.02296	[-0.0582, 0.0123]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.01084	[-0.0298, 0.00815]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.00556	[-0.0199, 0.00875]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.01186	[-0.0376, 0.0139]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.02826	[-0.0851, 0.0286]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.00694	[-0.0184, 0.00453]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.02628	[-0.0649, 0.0123]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.01276	[-0.0462, 0.0717]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.01492	[-0.0617, 0.0319]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01814	[-0.0517, 0.0154]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.01134	[-0.0411, 0.0184]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.00466	[-0.0417, 0.0324]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00934	[-0.0418, 0.0231]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.0189	[-0.0566, 0.0188]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.02048	[-0.0388, -0.00218]	0.5754
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.01458	[-0.0339, 0.0631]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.00872	[-0.0442, 0.0267]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.0076	[-0.0278, 0.0126]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.00546	[-0.038, 0.027]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.01224	[-0.0465, 0.022]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.0101	[-0.0401, 0.0199]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.00316	[-0.0356, 0.0292]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.0106	[-0.0112, 0.0324]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.02164	[-0.491, 0.534]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.1836	[-0.412, 0.779]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.3224	[-0.21, 0.855]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.0079	[-0.749, 0.733]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.1791	[-0.654, 0.295]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.07928	[-0.564, 0.723]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.1334	[-0.194, 0.461]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.08582	[-0.142, 0.314]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-1.400×10^{-4}	[-0.0278, 0.0275]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-6.800×10^{-4}	[-0.0459, 0.0446]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.0149	[-0.0254, -0.00435]	0.4130
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.01124	[-0.0243, 0.00187]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.00504	[-0.0259, 0.0158]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.00882	[-0.0465, 0.0642]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.01512	[-0.0573, 0.0271]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.0164	[-0.0382, 0.071]	1.0000

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9976	[0.997, 0.998]	5.85×10^{-4}	5	0
adaptive	pointer	0.01	0.9977	[0.997, 0.998]	7.63×10^{-4}	5	0
adaptive	pointer	0.1	0.9977	[0.997, 0.998]	7.46×10^{-4}	5	0
adaptive	pointer	1	0.9977	[0.997, 0.998]	7.78×10^{-4}	5	0
adaptive	pointer	10	0.9977	[0.997, 0.998]	7.32×10^{-4}	5	0
adaptive	pointer	100	0.9977	[0.997, 0.998]	7.34×10^{-4}	5	0
adaptive	pointer	1000	0.9976	[0.997, 0.998]	7.40×10^{-4}	5	0
adaptive	pointer	1e+04	0.9977	[0.997, 0.998]	7.71×10^{-4}	5	0
adaptive	seq2seq	0	0.9977	[0.997, 0.998]	4.47×10^{-4}	5	0
adaptive	seq2seq	0.01	0.9981	[0.998, 0.998]	4.71×10^{-4}	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.9981	[0.998, 0.998]	4.73×10^{-4}	5	0
adaptive	seq2seq	1	0.9982	[0.998, 0.998]	4.36×10^{-4}	5	0
adaptive	seq2seq	10	0.9982	[0.998, 0.998]	4.45×10^{-4}	5	0
adaptive	seq2seq	100	0.9982	[0.998, 0.999]	4.75×10^{-4}	5	0
adaptive	seq2seq	1000	0.9983	[0.998, 0.999]	4.59×10^{-4}	5	0
adaptive	seq2seq	1e+04	0.9981	[0.998, 0.998]	4.99×10^{-4}	5	0
tropical	pointer	0	0.9975	[0.997, 0.999]	0.00127	5	0
tropical	pointer	0.01	0.9977	[0.997, 0.999]	0.00148	5	0
tropical	pointer	0.1	0.9974	[0.997, 0.998]	8.13×10^{-4}	5	0
tropical	pointer	1	0.9976	[0.997, 0.998]	9.58×10^{-4}	5	0
tropical	pointer	10	0.9979	[0.997, 0.999]	0.00112	5	0
tropical	pointer	100	0.9975	[0.997, 0.998]	0.00107	5	0
tropical	pointer	1000	0.9979	[0.997, 0.999]	0.00123	5	0
tropical	pointer	1e+04	0.9977	[0.997, 0.999]	0.00141	5	0
tropical	seq2seq	0	0.9986	[0.997, 0.999]	0.00136	5	0
tropical	seq2seq	0.01	0.9987	[0.998, 0.999]	0.00104	5	0
tropical	seq2seq	0.1	0.9986	[0.998, 0.999]	0.00106	5	0
tropical	seq2seq	1	0.9986	[0.997, 0.999]	0.00138	5	0
tropical	seq2seq	10	0.9983	[0.997, 0.999]	0.00153	5	0
tropical	seq2seq	100	0.9984	[0.998, 0.999]	0.00109	5	0
tropical	seq2seq	1000	0.9989	[0.998, 0.999]	6.70×10^{-4}	5	0
tropical	seq2seq	1e+04	0.9988	[0.998, 0.999]	9.64×10^{-4}	5	0
vanilla	pointer	0	0.9975	[0.997, 0.998]	3.23×10^{-4}	5	0
vanilla	pointer	0.01	0.9978	[0.997, 0.998]	6.56×10^{-4}	5	0
vanilla	pointer	0.1	0.9978	[0.997, 0.998]	6.89×10^{-4}	5	0
vanilla	pointer	1	0.9978	[0.997, 0.998]	6.48×10^{-4}	5	0
vanilla	pointer	10	0.9977	[0.997, 0.998]	6.57×10^{-4}	5	0
vanilla	pointer	100	0.9978	[0.997, 0.998]	6.58×10^{-4}	5	0
vanilla	pointer	1000	0.9978	[0.997, 0.998]	6.51×10^{-4}	5	0
vanilla	pointer	1e+04	0.9978	[0.997, 0.998]	6.86×10^{-4}	5	0
vanilla	seq2seq	0	0.998	[0.998, 0.998]	3.13×10^{-4}	5	0
vanilla	seq2seq	0.01	0.9984	[0.998, 0.999]	2.86×10^{-4}	5	0
vanilla	seq2seq	0.1	0.9984	[0.998, 0.999]	2.56×10^{-4}	5	0
vanilla	seq2seq	1	0.9984	[0.998, 0.999]	3.06×10^{-4}	5	0
vanilla	seq2seq	10	0.9984	[0.998, 0.999]	2.41×10^{-4}	5	0
vanilla	seq2seq	100	0.9984	[0.998, 0.999]	2.33×10^{-4}	5	0
vanilla	seq2seq	1000	0.9984	[0.998, 0.999]	2.52×10^{-4}	5	0
vanilla	seq2seq	1e+04	0.9983	[0.998, 0.999]	2.79×10^{-4}	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	8.380×10^{-6}	$[-1.58 \times 10^{-4}, 3.27 \times 10^{-4}]$	1.0000
adaptive	pointer	0.1	-8.239×10^{-6}	$[-1.48 \times 10^{-4}, 3.11 \times 10^{-4}]$	1.0000
adaptive	pointer	1	8.156×10^{-6}	$[-1.75 \times 10^{-4}, 2.84 \times 10^{-4}]$	1.0000
adaptive	pointer	10	2.812×10^{-8}	$[-1.43 \times 10^{-4}, 3.39 \times 10^{-4}]$	1.0000
adaptive	pointer	100	-4.137×10^{-5}	$[-1.51 \times 10^{-4}, 2.72 \times 10^{-4}]$	1.0000
adaptive	pointer	1000	-2.489×10^{-5}	$[-1.76 \times 10^{-4}, 2.24 \times 10^{-4}]$	1.0000
adaptive	pointer	1e+04	-1.654×10^{-5}	$[-1.60 \times 10^{-4}, 2.74 \times 10^{-4}]$	1.0000
adaptive	seq2seq	0.01	5.550×10^{-4}	$[2.98 \times 10^{-4}, 6.42 \times 10^{-4}]$	0.4375
adaptive	seq2seq	0.1	5.300×10^{-4}	$[3.45 \times 10^{-4}, 6.46 \times 10^{-4}]$	0.4375
adaptive	seq2seq	1	5.300×10^{-4}	$[3.62 \times 10^{-4}, 6.45 \times 10^{-4}]$	0.4375
adaptive	seq2seq	10	4.887×10^{-4}	$[3.71 \times 10^{-4}, 6.98 \times 10^{-4}]$	0.4375

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	seq2seq	100	6.051×10^{-4}	$[4.07 \times 10^{-4}, 7.45 \times 10^{-4}]$	0.4375
adaptive	seq2seq	1000	7.210×10^{-4}	$[3.86 \times 10^{-4}, 8.21 \times 10^{-4}]$	0.4375
adaptive	seq2seq	1e+04	2.164×10^{-4}	$[1.33 \times 10^{-4}, 8.24 \times 10^{-4}]$	0.4375
tropical	pointer	0.01	-7.455×10^{-4}	$[-8.14 \times 10^{-4}, 0.0017]$	1.0000
tropical	pointer	0.1	-3.577×10^{-4}	$[-8.21 \times 10^{-4}, 6.32 \times 10^{-4}]$	1.0000
tropical	pointer	1	1.756×10^{-4}	$[-9.37 \times 10^{-4}, 9.52 \times 10^{-4}]$	1.0000
tropical	pointer	10	4.259×10^{-4}	$[-8.69 \times 10^{-4}, 0.00145]$	1.0000
tropical	pointer	100	-1.161×10^{-4}	$[-8.18 \times 10^{-4}, 7.36 \times 10^{-4}]$	1.0000
tropical	pointer	1000	5.794×10^{-5}	$[-6.61 \times 10^{-4}, 0.00135]$	1.0000
tropical	pointer	1e+04	3.333×10^{-5}	[-0.00124, 0.00169]	1.0000
tropical	seq2seq	0.01	8.283×10^{-6}	$[-4.96 \times 10^{-4}, 6.08 \times 10^{-4}]$	1.0000
tropical	seq2seq	0.1	-8.276×10^{-6}	$[-7.44 \times 10^{-4}, 5.77 \times 10^{-4}]$	1.0000
tropical	seq2seq	1	0	$[-3.19 \times 10^{-4}, 7.11 \times 10^{-5}]$	1.0000
tropical	seq2seq	10	0	$[-0.001, 8.60 \times 10^{-5}]$	1.0000
tropical	seq2seq	100	-1.159×10^{-4}	$[-9.23 \times 10^{-4}, 5.29 \times 10^{-4}]$	1.0000
tropical	seq2seq	1000	8.268×10^{-6}	$[-2.21 \times 10^{-4}, 9.04 \times 10^{-4}]$	1.0000
tropical	seq2seq	1e+04	8.268×10^{-6}	$[-2.81 \times 10^{-5}, 5.75 \times 10^{-4}]$	1.0000
vanilla	pointer	0.01	1.162×10^{-4}	$[-7.21 \times 10^{-6}, 5.06 \times 10^{-4}]$	0.8750
vanilla	pointer	0.1	4.961×10^{-5}	$[1.28 \times 10^{-5}, 5.74 \times 10^{-4}]$	0.8750
vanilla	pointer	1	1.578×10^{-4}	$[3.78 \times 10^{-5}, 5.39 \times 10^{-4}]$	0.8750
vanilla	pointer	10	7.419×10^{-5}	$[-1.87 \times 10^{-5}, 4.99 \times 10^{-4}]$	0.8750
vanilla	pointer	100	1.161×10^{-4}	$[7.86 \times 10^{-6}, 5.27 \times 10^{-4}]$	0.8750
vanilla	pointer	1000	1.079×10^{-4}	$[2.76 \times 10^{-5}, 5.16 \times 10^{-4}]$	0.8750
vanilla	pointer	1e+04	1.329×10^{-4}	$[7.99 \times 10^{-6}, 5.56 \times 10^{-4}]$	0.8750
vanilla	seq2seq	0.01	3.394×10^{-4}	$[1.74 \times 10^{-4}, 6.00 \times 10^{-4}]$	0.4375
vanilla	seq2seq	0.1	2.730×10^{-4}	$[1.19 \times 10^{-4}, 5.56 \times 10^{-4}]$	0.4375
vanilla	seq2seq	1	1.573×10^{-4}	$[1.21 \times 10^{-4}, 5.92 \times 10^{-4}]$	0.4375
vanilla	seq2seq	10	3.644×10^{-4}	$[1.39 \times 10^{-4}, 5.93 \times 10^{-4}]$	0.4375
vanilla	seq2seq	100	2.978×10^{-4}	$[9.40 \times 10^{-5}, 5.81 \times 10^{-4}]$	0.4375
vanilla	seq2seq	1000	3.973×10^{-4}	$[1.52 \times 10^{-4}, 5.53 \times 10^{-4}]$	0.4375
vanilla	seq2seq	1e+04	1.737×10^{-4}	$[5.12 \times 10^{-5}, 4.87 \times 10^{-4}]$	0.4375

Table 27: `id_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.1315	[-0.439, 0.111]	0.4513	-0.0971	0.4673
adaptive	seq2seq	-0.01839	[-0.143, 0.289]	0.9165	-0.00571	0.9764
tropical	pointer	0.08063	[-0.232, 0.318]	0.6452	0.0743	0.5935
tropical	seq2seq	-0.01346	[-0.19, 0.139]	0.9388	0.00381	0.9826
vanilla	pointer	-0.008487	[-0.221, 0.106]	0.9614	-0.0019	1.0000
vanilla	seq2seq	-0.1514	[-0.429, 0.0113]	0.3854	-0.122	0.3739

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	1.428×10^{-5}	[-0.0015, 0.00152]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-5.791×10^{-5}	[-0.00122, 0.00111]	1.0000

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-3.987×10^{-4}	$[-9.28 \times 10^{-4}, 1.31 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-2.349×10^{-4}	$[-7.43 \times 10^{-4}, 2.73 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	1.226×10^{-4}	[-0.00107, 0.00131]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-2.883×10^{-4}	$[-0.00116, 5.88 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	9.725×10^{-5}	$[-9.56 \times 10^{-4}, 0.00115]$	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-1.159×10^{-4}	$[-0.00109, 8.57 \times 10^{-4}]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	5.912×10^{-4}	$[-9.90 \times 10^{-4}, 0.00217]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	2.643×10^{-4}	$[-7.48 \times 10^{-4}, 0.00128]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	2.181×10^{-4}	[-0.00101, 0.00145]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	1.538×10^{-4}	[-0.00127, 0.00158]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-7.551×10^{-5}	[-0.00175, 0.00159]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	7.341×10^{-5}	[-0.00101, 0.00116]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	4.685×10^{-4}	$[-2.41 \times 10^{-4}, 0.00118]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	5.478×10^{-4}	$[-3.58 \times 10^{-4}, 0.00145]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	1.276×10^{-4}	$[-3.13 \times 10^{-4}, 5.68 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-5.955×10^{-5}	$[-3.24 \times 10^{-4}, 2.05 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-8.782×10^{-5}	$[-2.67 \times 10^{-4}, 9.18 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-1.295×10^{-4}	$[-4.36 \times 10^{-4}, 1.77 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-2.815×10^{-5}	$[-2.75 \times 10^{-4}, 2.18 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-9.111×10^{-5}	$[-3.02 \times 10^{-4}, 1.19 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-1.409×10^{-4}	$[-4.20 \times 10^{-4}, 1.38 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-1.211×10^{-4}	$[-4.16 \times 10^{-4}, 1.74 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-3.922×10^{-4}	$[-9.82 \times 10^{-4}, 1.98 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-3.000×10^{-4}	$[-5.43 \times 10^{-4}, -5.75 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-2.271×10^{-4}	$[-5.04 \times 10^{-4}, 4.96 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-2.405×10^{-4}	$[-4.19 \times 10^{-4}, -6.15 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-2.206×10^{-4}	$[-4.85 \times 10^{-4}, 4.40 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-1.409×10^{-4}	$[-4.72 \times 10^{-4}, 1.91 \times 10^{-4}]$	1.0000

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-1.359×10^{-4}	$[-4.01 \times 10^{-4}, 1.29 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-1.840×10^{-4}	$[-7.56 \times 10^{-4}, 3.88 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-2.552×10^{-5}	$[-4.25 \times 10^{-4}, 3.74 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-4.355×10^{-4}	$[-0.00112, 2.45 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-4.586×10^{-4}	$[-0.00116, 2.39 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-5.019×10^{-4}	$[-0.00126, 2.55 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-4.753×10^{-4}	$[-0.00121, 2.61 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-5.582×10^{-4}	$[-0.00109, -2.33 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-6.263×10^{-4}	$[-0.00133, 7.97 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-4.540×10^{-4}	$[-0.00149, 5.79 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.001122	[-0.00388, 0.00163]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-9.981×10^{-4}	$[-0.00271, 7.16 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.001215	$[-0.00242, -6.50 \times 10^{-6}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.001002	$[-0.00231, 3.03 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-4.697×10^{-4}	$[-0.00153, 5.91 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-9.696×10^{-4}	$[-0.00244, 5.00 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-9.924×10^{-4}	$[-0.00204, 5.43 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.001181	$[-0.00247, 1.04 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-5.453×10^{-4}	$[-9.11 \times 10^{-4}, -1.80 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-6.759×10^{-4}	$[-0.00124, -1.10 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-5.979×10^{-4}	$[-0.00124, 4.56 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-6.129×10^{-4}	$[-0.00117, -5.32 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-6.678×10^{-4}	$[-0.00128, -5.73 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-6.079×10^{-4}	$[-0.00124, 2.80 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-6.212×10^{-4}	$[-0.00118, -6.51 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-5.169×10^{-4}	$[-0.00112, 8.26 \times 10^{-5}]$	1.0000

A.7 id_exact

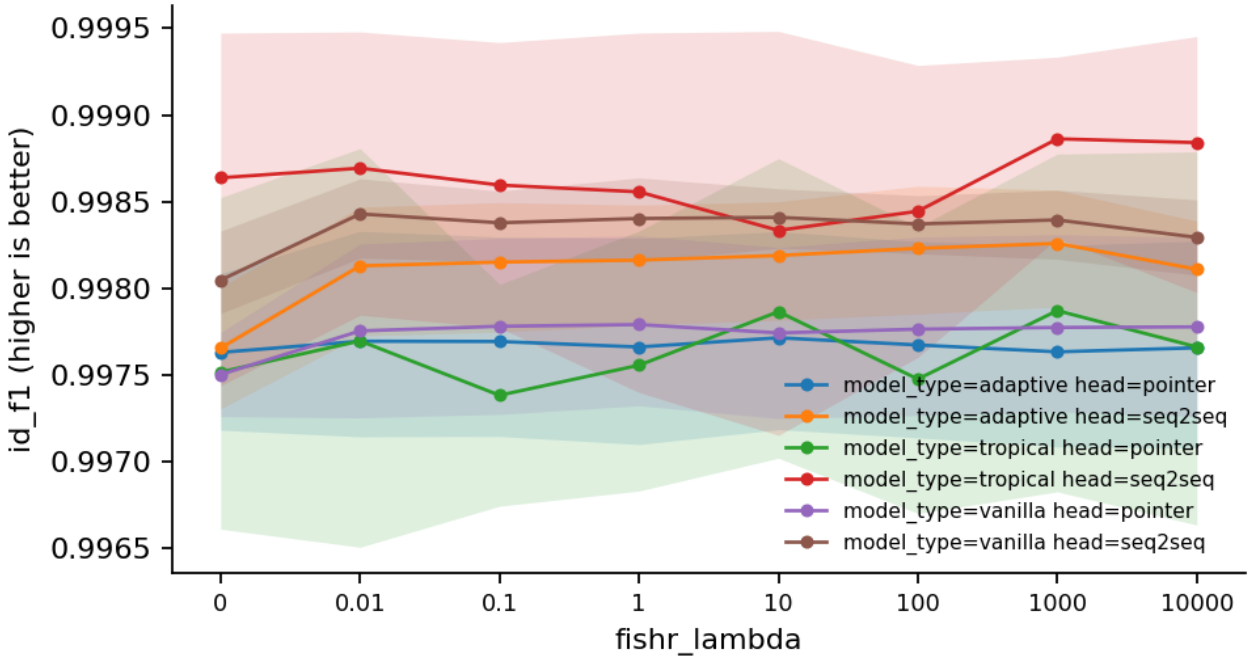


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

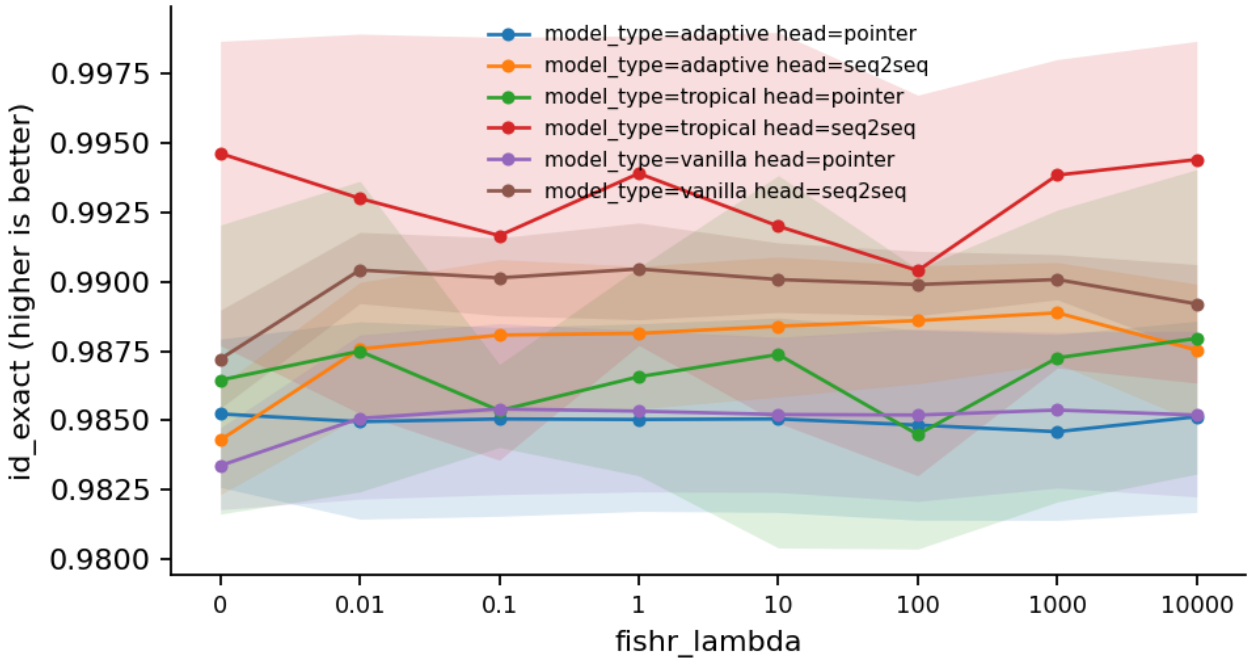


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9852	[0.983, 0.988]	0.00349	5	0
adaptive	pointer	0.01	0.985	[0.981, 0.989]	0.00456	5	0
adaptive	pointer	0.1	0.9851	[0.982, 0.988]	0.00441	5	0
adaptive	pointer	1	0.985	[0.982, 0.988]	0.00455	5	0
adaptive	pointer	10	0.9851	[0.982, 0.989]	0.00423	5	0
adaptive	pointer	100	0.9848	[0.981, 0.988]	0.00434	5	0
adaptive	pointer	1000	0.9846	[0.981, 0.988]	0.00442	5	0
adaptive	pointer	1e+04	0.9851	[0.982, 0.989]	0.00433	5	0
adaptive	seq2seq	0	0.9843	[0.982, 0.986]	0.00248	5	0
adaptive	seq2seq	0.01	0.9876	[0.985, 0.99]	0.0031	5	0
adaptive	seq2seq	0.1	0.9881	[0.985, 0.991]	0.00356	5	0
adaptive	seq2seq	1	0.9881	[0.985, 0.991]	0.00332	5	0
adaptive	seq2seq	10	0.9884	[0.986, 0.991]	0.00319	5	0
adaptive	seq2seq	100	0.9886	[0.986, 0.991]	0.00282	5	0
adaptive	seq2seq	1000	0.9889	[0.987, 0.991]	0.00241	5	0
adaptive	seq2seq	1e+04	0.9875	[0.985, 0.99]	0.00324	5	0
tropical	pointer	0	0.9865	[0.982, 0.992]	0.00664	5	0
tropical	pointer	0.01	0.9875	[0.982, 0.994]	0.00718	5	0
tropical	pointer	0.1	0.9854	[0.984, 0.987]	0.0019	5	0
tropical	pointer	1	0.9866	[0.983, 0.991]	0.00493	5	0
tropical	pointer	10	0.9874	[0.98, 0.994]	0.00908	5	0
tropical	pointer	100	0.9845	[0.98, 0.99]	0.0066	5	0
tropical	pointer	1000	0.9873	[0.982, 0.993]	0.007	5	0
tropical	pointer	1e+04	0.988	[0.983, 0.994]	0.00719	5	0
tropical	seq2seq	0	0.9946	[0.988, 0.999]	0.00779	5	0
tropical	seq2seq	0.01	0.993	[0.985, 0.999]	0.00874	5	0
tropical	seq2seq	0.1	0.9917	[0.984, 0.999]	0.00977	5	0
tropical	seq2seq	1	0.9939	[0.988, 0.999]	0.00782	5	0
tropical	seq2seq	10	0.992	[0.985, 0.999]	0.00939	5	0
tropical	seq2seq	100	0.9904	[0.983, 0.997]	0.00894	5	0
tropical	seq2seq	1000	0.9938	[0.987, 0.998]	0.00759	5	0
tropical	seq2seq	1e+04	0.9944	[0.986, 0.999]	0.00895	5	0
vanilla	pointer	0	0.9834	[0.982, 0.985]	0.00178	5	0
vanilla	pointer	0.01	0.9851	[0.982, 0.988]	0.00395	5	0
vanilla	pointer	0.1	0.9854	[0.982, 0.988]	0.00402	5	0
vanilla	pointer	1	0.9853	[0.982, 0.988]	0.00378	5	0
vanilla	pointer	10	0.9852	[0.982, 0.988]	0.00369	5	0
vanilla	pointer	100	0.9852	[0.982, 0.988]	0.00397	5	0
vanilla	pointer	1000	0.9854	[0.983, 0.988]	0.00373	5	0
vanilla	pointer	1e+04	0.9852	[0.982, 0.988]	0.004	5	0
vanilla	seq2seq	0	0.9872	[0.985, 0.989]	0.00231	5	0
vanilla	seq2seq	0.01	0.9904	[0.989, 0.992]	0.00154	5	0
vanilla	seq2seq	0.1	0.9901	[0.989, 0.992]	0.0018	5	0
vanilla	seq2seq	1	0.9905	[0.989, 0.992]	0.00225	5	0
vanilla	seq2seq	10	0.9901	[0.989, 0.991]	0.00169	5	0
vanilla	seq2seq	100	0.9899	[0.989, 0.991]	0.00151	5	0
vanilla	seq2seq	1000	0.9901	[0.989, 0.991]	0.00105	5	0
vanilla	seq2seq	1e+04	0.9892	[0.987, 0.991]	0.00209	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-9.000×10^{-4}	[-0.00146, 0.00124]	1.0000
adaptive	pointer	0.1	-6.000×10^{-4}	[-0.00144, 0.00112]	1.0000
adaptive	pointer	1	-7.000×10^{-4}	[-0.00138, 0.00102]	1.0000
adaptive	pointer	10	-5.000×10^{-4}	[-0.00124, 0.00108]	1.0000
adaptive	pointer	100	-4.000×10^{-4}	[-0.00144, 9.00×10^{-4}]	1.0000
adaptive	pointer	1000	-6.000×10^{-4}	[-0.0018, 5.20×10^{-4}]	1.0000
adaptive	pointer	1e+04	0	[-0.00128, 0.00114]	1.0000
adaptive	seq2seq	0.01	0.0028	[0.00188, 0.00468]	0.4375
adaptive	seq2seq	0.1	0.0036	[0.00228, 0.00526]	0.4375
adaptive	seq2seq	1	0.0029	[0.00258, 0.00518]	0.4375
adaptive	seq2seq	10	0.0039	[0.00268, 0.00532]	0.4375
adaptive	seq2seq	100	0.0035	[0.00286, 0.00594]	0.4375
adaptive	seq2seq	1000	0.0052	[0.00284, 0.00618]	0.4375
adaptive	seq2seq	1e+04	0.0018	[6.00×10^{-4}, 0.00594]	0.4375
tropical	pointer	0.01	-5.000×10^{-4}	[-0.0052, 0.00958]	1.0000
tropical	pointer	0.1	0.0011	[-0.0053, 0.00248]	1.0000
tropical	pointer	1	3.000×10^{-4}	[-0.00538, 0.0051]	1.0000
tropical	pointer	10	0.0019	[-0.00926, 0.00966]	1.0000
tropical	pointer	100	-4.000×10^{-4}	[-0.00718, 0.0018]	1.0000
tropical	pointer	1000	4.000×10^{-4}	[-0.006, 0.00644]	1.0000
tropical	pointer	1e+04	0	[-0.00643, 0.0103]	1.0000
tropical	seq2seq	0.01	-1.000×10^{-4}	[-0.00514, 8.80×10^{-4}]	1.0000
tropical	seq2seq	0.1	-3.000×10^{-4}	[-0.00848, 7.40×10^{-4}]	1.0000
tropical	seq2seq	1	0	[-0.00334, 0.00114]	1.0000
tropical	seq2seq	10	0	[-0.00922, 0.00132]	1.0000
tropical	seq2seq	100	-0.0025	[-0.0097, -2.40×10^{-4}]	1.0000
tropical	seq2seq	1000	-3.000×10^{-4}	[-0.0027, 9.40×10^{-4}]	1.0000
tropical	seq2seq	1e+04	0	[-0.00148, 0.00114]	1.0000
vanilla	pointer	0.01	4.000×10^{-4}	[-2.00×10^{-4}, 0.0036]	0.9375
vanilla	pointer	0.1	6.000×10^{-4}	[1.60×10^{-4}, 0.00392]	0.8750
vanilla	pointer	1	7.000×10^{-4}	[2.20×10^{-4}, 0.00378]	0.8750
vanilla	pointer	10	7.000×10^{-4}	[1.40×10^{-4}, 0.00354]	0.8750
vanilla	pointer	100	4.000×10^{-4}	[-2.00×10^{-5}, 0.00368]	0.9375
vanilla	pointer	1000	0.001	[2.40×10^{-4}, 0.00376]	0.8750
vanilla	pointer	1e+04	6.000×10^{-4}	[-1.20×10^{-4}, 0.00376]	0.9375
vanilla	seq2seq	0.01	0.0037	[0.0016, 0.00482]	0.4375
vanilla	seq2seq	0.1	0.0035	[0.00102, 0.0047]	0.5000
vanilla	seq2seq	1	0.0025	[0.0013, 0.00538]	0.4375
vanilla	seq2seq	10	0.0035	[1.00×10^{-3}, 0.00458]	0.5000
vanilla	seq2seq	100	0.0034	[9.20×10^{-4}, 0.00444]	0.5000
vanilla	seq2seq	1000	0.0032	[0.00116, 0.00456]	0.4375
vanilla	seq2seq	1e+04	0.0017	[6.00×10^{-4}, 0.00356]	0.5000

Table 31: `id_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.009203	[-0.206, 0.122]	0.9582	0.00762	0.9626
adaptive	seq2seq	0.06296	[-0.191, 0.381]	0.7194	0.0324	0.8204
tropical	pointer	-0.01344	[-0.359, 0.256]	0.9389	0	1.0000
tropical	seq2seq	-0.04608	[-0.304, 0.234]	0.7926	-0.0248	0.8542
vanilla	pointer	0.01417	[-0.156, 0.144]	0.9356	0.00952	0.9478
vanilla	seq2seq	-0.1382	[-0.38, -0.0639]	0.4286	-0.107	0.4231

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.00308	[-0.00557, 0.0117]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.00242	[-0.00336, 0.0082]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-6.000×10^{-5}	[-0.00473, 0.00461]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.00124	[-0.00162, 0.0041]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.00216	[-0.00836, 0.0127]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-7.000×10^{-4}	[-0.0061, 0.0047]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.00188	[-0.00548, 0.00924]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.00276	[-0.00266, 0.00818]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.0074	$[-5.70 \times 10^{-4}$, 0.0154]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.00258	[-0.00695, 0.0121]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.00152	[-0.0109, 0.0139]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.00344	[-0.00425, 0.0111]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.00192	[-0.00905, 0.0129]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	5.000×10^{-4}	[-0.00943, 0.0104]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.00376	[-0.00479, 0.0123]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.0052	[-0.00366, 0.0141]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00186	$[-4.25 \times 10^{-4}$, 0.00415]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-1.200×10^{-4}	[-0.00191, 0.00167]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-3.600×10^{-4}	[-0.00146, 7.39×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-3.000×10^{-4}	[-0.00187, 0.00127]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-1.600×10^{-4}	[-0.00181, 0.00149]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-3.600×10^{-4}	[-0.00157, 8.46×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-7.800×10^{-4}	[-0.00243, 8.68×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-6.000×10^{-5}	[-0.00183, 0.00171]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.00292	[-0.00724, 0.0014]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.00284	[-0.00545, -2.29×10^{-4}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00206	[-0.00533, 0.00121]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.00232	[-0.00401, -6.30×10^{-4}]	1.0000

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.00168	[-0.00462, 0.00126]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0013	[-0.00395, 0.00135]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.0012	[-0.00343, 0.00103]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00166	[-0.00648, 0.00316]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	9.400×10^{-4}	[-0.00286, 0.00474]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.00262	[-0.00805, 0.00281]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.00302	[-0.00868, 0.00264]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.0031	[-0.00876, 0.00256]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.00334	[-0.00914, 0.00246]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.00376	[-0.00813, 6.08×10^{-4}]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.00428	[-0.00956, 0.001]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.0024	[-0.0106, 0.00584]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.00816	[-0.0235, 0.00713]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.0055	[-0.0166, 0.00557]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.0063	[-0.0184, 0.00576]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.00732	[-0.0134, -0.00124]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.00462	[-0.0117, 0.00241]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.0059	[-0.0146, 0.00277]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.00658	[-0.0129, -2.21×10^{-4}]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.00644	[-0.0163, 0.00338]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.00384	[-0.00683, -8.51×10^{-4}]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.00534	[-0.0104, -2.40×10^{-4}]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.00472	[-0.0106, 0.00112]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.00512	[-0.00991, -3.35×10^{-4}]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.00486	[-0.0102, 4.63×10^{-4}]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.0047	[-0.0101, 6.94×10^{-4}]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.0047	[-0.00958, 1.77×10^{-4}]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.004	[-0.00921, 0.00121]	1.0000

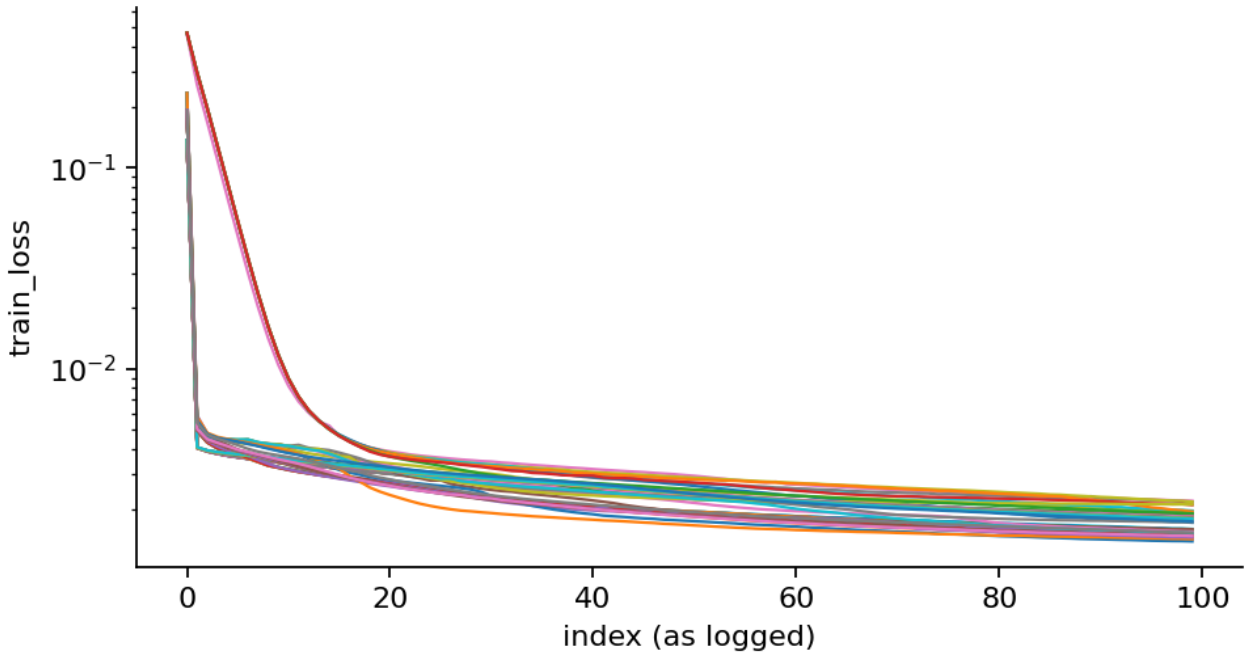


Figure 9: `train_loss`, the mean over seeds per cell.

B Curves

C Configuration

The base configuration every run shares (frozen hash `0ea3d4b19330`); factor levels are in Table 1.

```
task:
  name: SCCDataset
  value_range: 0.001
  target_range: null
  weight_range: null
  adversarial_range: null
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  ood_value_range: 0.1
  ood_noise_range: null
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
```

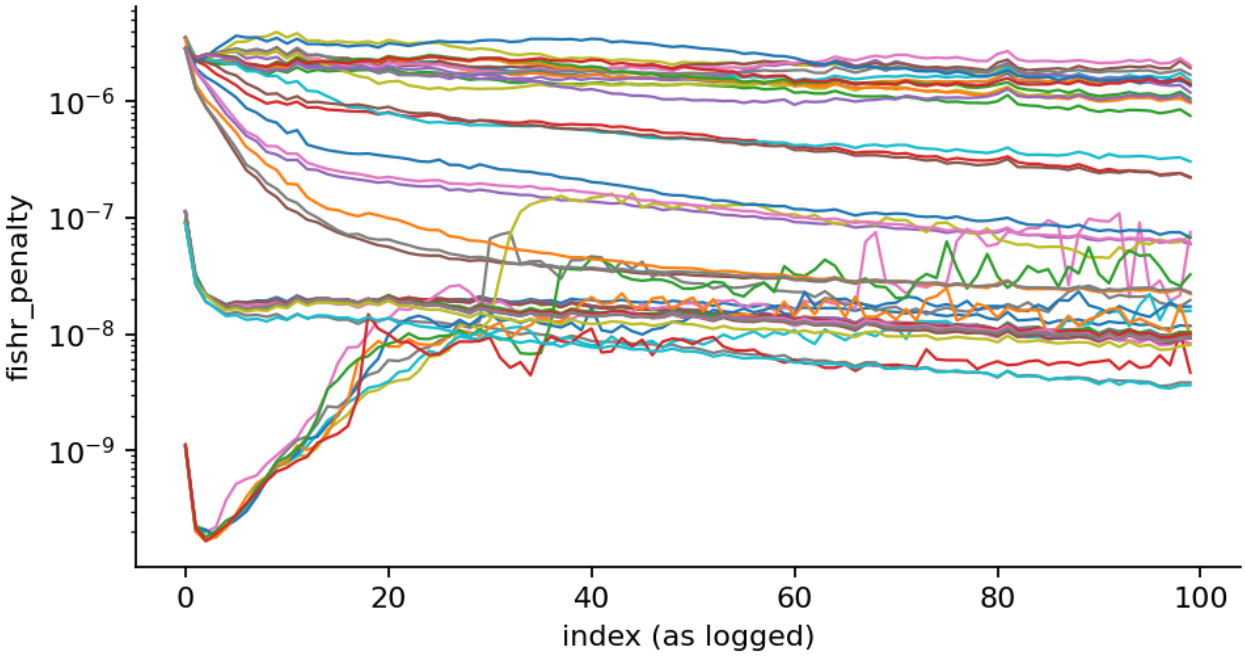


Figure 10: `fishr_penalty`, the mean over seeds per cell.

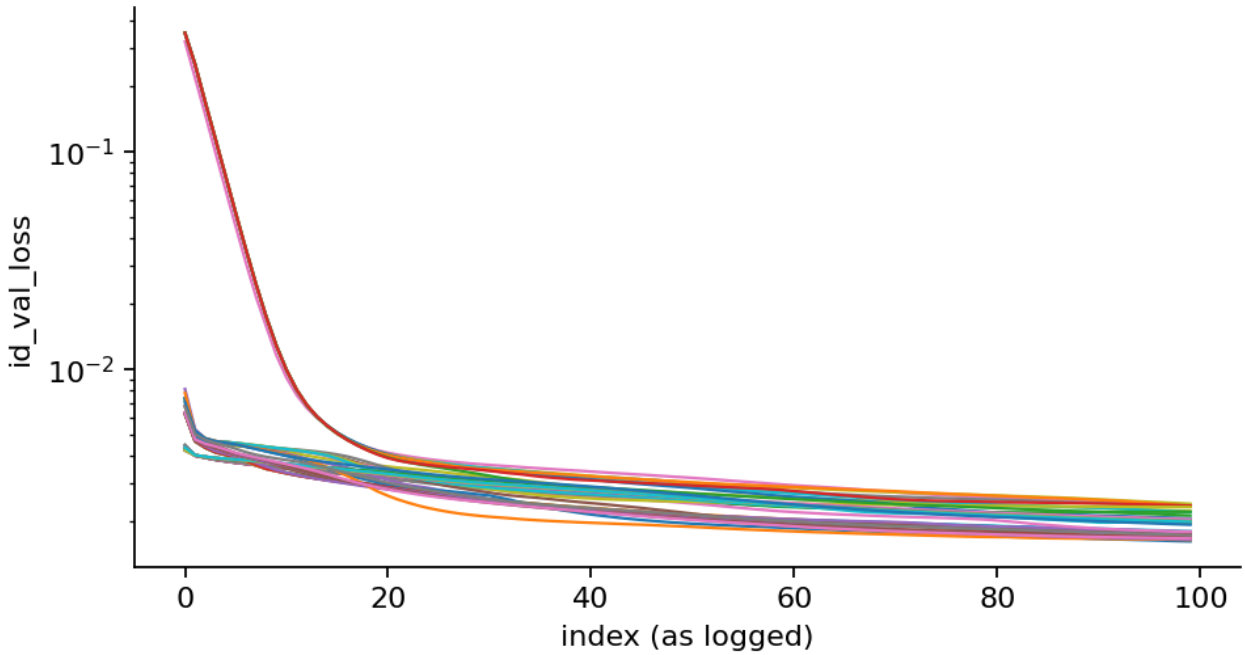


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  lr: 0.0001
  epochs: 100
  batch_size: 8
  eval_batch_size: 8
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/SCC/results`.